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Research Report

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A U.S. Strategy to Prevent the Creation of Mirror Life

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About This Report

Advances in synthetic biology and molecular engineering could lead to the creation of *mirror life*, a hypothetical form of life composed of molecules with reversed chirality relative to all known life on earth. In December 2024, a group of concerned scientists warned about the potential risks of mirror life. Its reversed molecular structure could enable mirror bacteria to escape immune detection and cause fatal infections, resist degradation, avoid natural predators, and proliferate through natural ecosystems. Even a small initial population of mirror organisms could potentially spread widely, causing irreversible ecological disruption. If created and released, whether accidentally or intentionally, mirror life could pose a catastrophic risk to multicellular life on earth, including humans, animals, and plants.

These risks raise important questions about the governance of research that could lead to mirror life, as well as broader strategic questions about how the United States and the world can safeguard against these potentially catastrophic outcomes.

In this report, we define the strategic challenges posed by the possibility of mirror life and examine potential levers to address those challenges. We propose a strategy that the United States could pursue to minimize the chance that mirror life is ever created by anyone, anywhere. The primary audiences for this report include science advisory bodies, regulatory and policy institutions, national security decisionmakers, and scientists whose work may intersect with the pathways to mirror life.

This report builds on prior RAND research on mirror life, including *Policy Options to Prevent the Creation of Mirror Organisms* (Epstein, Crawford, and Nevo, 2025), as well as a body of research on biosecurity, dual-use technologies, and emerging technology governance. The report contributes to a broader institutional effort to understand and manage transformative scientific risks.

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Summary

Mirror life is a hypothetical form of life built from biomolecules with the opposite chirality—or molecular handedness—of all known life on earth. Although mirror life does not yet exist, experts estimate that creation of the first mirror bacterium—a likely first instantiation of mirror life—could be technically feasible within the next few decades and could potentially cost as little as several hundred million dollars.

Its reversed molecular structure could enable mirror bacteria to escape immune detection and cause fatal infections, resist degradation, avoid natural predators, and proliferate throughout natural ecosystems. Even a small initial population of mirror organisms could potentially spread widely, causing irreversible ecological disruption. If created and released, whether accidentally or intentionally, mirror life could pose a catastrophic risk to multicellular life on earth, including humans, animals, and plants.

The policy challenge is both urgent and unusual. Scientists have produced mirror-image biomolecules and other components that could serve as the building blocks of mirror life. Much of the research that moves science closer to realizing mirror organisms is not inherently dangerous, has demonstrated benefits or promises benefits, and is likely to continue. However, mirror life itself is expected to have few benefits. Despite this, some actors might still wish to pursue mirror life, such as well-intentioned researchers who overlook the risks of mirror life or malicious actors who seek to acquire or use mirror life as a doomsday capability.

Traditional governance approaches for emerging technologies—such as post-hoc regulation or adaptive oversight—rely on the opportunity to mitigate risks as they emerge and before they become crises. Thus, they may be ill suited to govern mirror life. This is because once a single viable mirror bacterium exists, the knowledge and methods for making riskier variants could proliferate rapidly, and even seemingly benign strains could be modified to be more dangerous. The first successful creation of mirror life could therefore represent an irreversible threshold: a point after which containment and control become far more difficult and perhaps even impossible.

In this report, we intend to guide U.S. policymakers in developing a preventive strategy to address the risks posed by mirror life before scientific progress or geopolitical incentives close the window for meaningful action. Our aim is not to predict whether mirror life will emerge or to assert with certainty that its effects will be catastrophic. Rather, we take seriously the expert assessment that mirror life could expose much of the earth's biosphere to unacceptable risks, and we treat this possibility as sufficient motivation for strategic planning now.

We focus on a U.S. strategy because U.S. policymakers are our primary audience and because the United States, as a scientific superpower and global hub for bioscience and biotechnology, will strongly shape the global trajectory of mirror life development. U.S. actions could either help establish global restraint or accelerate a competitive race. Although multilateral and United Nations (UN)-based approaches may play important roles, their effectiveness will depend on U.S. engagement.

To explore how such a strategy could be constructed, we

- identify and categorize the actors who might pursue mirror life and their potential motivations and capabilities
- catalog a broad set of policy levers, including incentives, regulations, diplomatic tools, verification mechanisms, and norm-building efforts
- examine how the scientific characteristics of mirror life and the potential geopolitical implications of its creation produce a distinct governance challenge, and derive principles for assembling policy levers into an effective strategy
- develop a U.S. strategy to prevent the creation of mirror life by selecting and organizing levers that ensure that no actor has both the motivation and the ability to create mirror life.

Finally, although this report focuses on the U.S. role in managing the development of mirror life, the prevention of mirror life cannot be achieved by any single country. The risk of mirror life is global, and the United States' most constructive role is as a partner in cooperation among the major scientific powers, particularly China. The result is a strategic framework that could help the United States and the world prevent the creation of mirror life.

Strategic Principles and Recommended Lines of Effort

Several features of mirror life make existing governance models inadequate and point to the following four principles that must guide any strategy to govern it:

1. **Prevention over mitigation.** Action must occur *before* the first mirror organism exists because the first successful experiment might also mark the point at which containment and defense become impossible.
2. **Universal restraint over partial restraint.** The creation of mirror life by *any* actor could pose a global catastrophic risk; thus, the motivations and capabilities of all actors must be considered, not just those of adversaries or competitors.
3. **Transparency over secrecy.** Given globally available materials and distributed scientific capabilities, secrecy is infeasible and destabilizing. Transparency enables trust and supports broad cooperation.
4. **Cooperation over coercion:** Preventing the creation of mirror life requires preserving the world's capacity for cooperation. Because only a few major scientific powers can create mirror life independently, their restraint is essential to global safety. This means prioritizing cooperation among these powers, including traditional competitors, and avoiding the use of tools that might erode trust.

Recommended Lines of Effort

To operationalize these principles, the United States should pursue the two lines of effort (LOEs) outlined in Table S.1:

- **LOE 1—Engage scientific superpowers, especially China, to build the foundation for collective restraint.** Prioritize transparent scientific collaboration, shared risk assessments, and early norm-setting so that no major power concludes that mirror life offers a strategic advantage and that mirror life does not become an arena for geopolitical competition.
- **LOE 2—Establish collective restraint across domestic, allied, and competing scientific communities.** Combine measures that reduce the incentives and increase the costs of creating mirror life. Together, these actions shift motivations and capabilities away from mirror life development among states, commercial firms, research institutions, and multilateral bodies. The collective refusal of scientifically advanced countries to pursue the science and engineering necessary for mirror life makes the pursuit of mirror life essentially impossible for remaining actors that are less advanced and dependent on development by others.

Supporting Recommendations

In pursuing these LOEs, the United States should also do the following:

- **Commit clearly to not developing mirror life,** even if other actors are suspected of pursuing it. This commitment reflects the fact that possession of mirror life would not provide reliable protection, deterrence, or resilience against mirror life developed by others and would instead exacerbate global risk.
- **Take great caution with actions or statements that could foster mistrust or spur strategic competition** in the field of mirror life or its enabling technologies, particularly actions or statements related to disrupting mirror life development or investing in countermeasures and resilience.
- **Advance actions with longer lead times in parallel.** Legislation, international engagement, technology development for detection and verification, and treaty groundwork require years to mature. These efforts should begin as soon as possible and evolve as new information emerges.
- **Exclude coercive measures from the strategy at this time.** As we discuss in Chapter 6, actions that rely on pressure, disruption, or punishment would erode the trust and cooperation among major scientific powers on which prevention depends. Such measures should be out of scope at this time. If such measures ever become appropriate, they should be pursued cooperatively and transparently, not unilaterally.
- **Treat countermeasures and resilience as constrained contingencies.** As we discuss in Chapter 6, these measures are unlikely to be effective against mirror life's risks, and premature or unilateral investment in defensive preparations creates the risks of weakening deterrence, advancing the science of mirror life, and triggering counterproductive dynamics.

Both LOEs advance the same goal: shaping a global environment in which restraint from creating mirror life becomes the shared norm among actors who are capable of creating it. By strengthening trust, transparency, and cooperative incentives, the United States can help ensure that those with the means to develop mirror life choose not to—and, in doing so, make mirror life's pursuit infeasible for everyone else.

TABLE S.1
Summary of a Possible U.S. Strategy for Preventing Mirror Life

	LOE 1: Engage Scientific Superpowers to Build the Foundation for Collective Restraint	LOE 2: Establish Collective Restraint
Objective	Shape early expectations by improving scientific understanding, aligning interpretations of risk among major powers, and establishing shared norms that make restraint the natural baseline.	Translate emerging consensus into durable domestic, bilateral, and multilateral constraints that raise the costs and lower the incentives of pursuing mirror life.
Levers implemented	• Conduct additional risk assessments to address remaining uncertainties. • Research safer alternatives to mirror life. • Develop voluntary and ethical standards. • Issue public and multilateral statements and resolutions.	• Detect and deter early-stage development. • Expand monitoring. • Develop verification methods. • Establish a global scientific oversight committee. • Increase direct economic barriers. • Restrict access to enabling materials, technologies, and knowledge. • Criminalize and prosecute domestic mirror life development.
Actors affected and unaffected	• Influences the incentives but not the capabilities of scientifically advanced states, academic institutions, commercial firms, and most nation-states • Has a limited effect on isolated and sanctioned states and little effect on extremist groups or lone individuals	• Shapes both incentives and capabilities of scientifically advanced states, commercial entities, and academic institutions • Constrains isolated states, sanctioned regimes, transnational criminal and violent extremist organizations, apocalyptic fringe groups, and lone individuals who cannot develop mirror life on their own

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CHAPTER 1

Introduction

In 2024, a group of scientists highlighted the potential for a scientifically plausible yet highly risky development in synthetic biology: the creation of mirror life. As described in Box 1.1, this hypothetical life form would be built from biomolecules with the opposite *chirality* (or handedness) of biomolecules found in all known life on earth. Whereas a natural-chirality cell contains "right-handed" nucleic acids and "left-handed" proteins, a mirror cell would have the reverse structure. Although mirror life does not yet exist, experts estimate that creation of the first mirror bacterium—a likely first instantiation of mirror life—could be technically feasible within the next few decades and could potentially cost as little as several hundred million dollars.

Some scientists warn that a reversed molecular structure could allow mirror bacteria to resist natural predators and immune defenses, enabling mirror life to proliferate widely and cause irreversible biological disruption. These scientists concluded that mirror life, if developed and released, could therefore pose a potentially catastrophic risk to most life on earth, including humans, animals, and plants (Adamala, Agashe, Belkaid, et al., 2024; Adamala, Agashe, Binder, et al., 2024).[1]

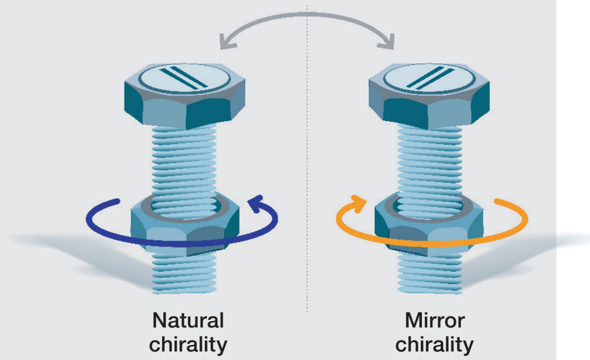
Despite the almost incomprehensible risks that might result from mirror life's creation, some actors may nonetheless pursue it. Some research programs have already made progress on precursor steps to synthesizing a mirror cell—for example, researchers have produced short mirror proteins and short mirror DNA sequences—but substantial barriers remain. Although these precursor developments could lead to novel tools for biomanufacturing or novel therapeutics, mirror life itself is expected to have few benefits (Adamala, Agashe, Binder, et al., 2024). Some actors might be motivated to pursue the creation of mirror life if those actors reject the scientific concerns about its dangers, believe that risks can be controlled, or assume mirror life could function as a controllable biological weapon. Others, including malicious actors, could be drawn to creating mirror life because it could be an uncontrollable doomsday device or provide leverage through the threat of one. However, attempting to develop mirror life as a biological weapon would likely be a grave strategic mistake for any actor interested in self-preservation: Mirror life could not be targeted at an enemy, would spread widely through ecosystems, and, over time, would threaten attacker and target alike (Adamala, Agashe, Binder, et al., 2024).

Although there is still significant uncertainty about the pathways to the creation of mirror life, experts estimate that mirror bacteria could be developed in a decade if actively pursued and potentially for as little as \$500 million (Adamala, Agashe, Belkaid, et al., 2024; Glass, 2025; Kubota, 2024). Without a focused and prioritized effort, timelines could stretch to 15 to 30 years, but ongoing advances in synthetic biology, artificial intelligence (AI), and chemical engineering could speed progress and drive down costs.

[1] Specifically, mirror life could pose serious risks to most multicellular organisms. According to Adamala, Agashe, Binder, et al., 2024, mirror organisms are unlikely to directly infect or consume single-celled organisms, such as bacteria, but they could still cause harm by competing for resources or disrupting ecological systems. For simplicity, this report refers to these risks collectively as threats to "life on earth," while recognizing that impacts would vary across different forms of life.

BOX 1.1

The Nuts and Bolts of Mirror Life



Biological molecules, or biomolecules, are the nuts and bolts of life on earth. Similar to real nuts and bolts, many of them share a fundamental handedness or chirality that shapes how they function. Nearly all nuts and bolts follow the familiar "righty-tighty, lefty-loosey" rule: They are threaded to tighten with turns to the right. The mirror image of the common right-handed bolt—a left-handed bolt—would be incompatible with a right-handed nut. This incompatibility illustrates chirality—structures that mirror each other but cannot be superimposed—much like left and right hands or left- and right-threaded nuts and bolts (Inaki, Liu, and Matsuno, 2016).

The biomolecules that compose all known life on earth exist only in one chirality. For instance, nucleic acids, such as DNA (deoxyribonucleic acid) and RNA (ribonucleic acid), are considered right-handed, and proteins are considered left-handed (Blackmond, 2019). Mirror biomolecules have the inverse orientation; mirror nucleic acids are left-handed, while mirror proteins are right-handed.

A living organism in which all biomolecules exist in their mirrored form would constitute mirror life. Although this could hypothetically include mirror squirrels or mirror nematodes, scientists are primarily concerned about mirror bacteria, which are perceived as the most plausible first instantiation of mirror life and the most dangerous, given their potential to be pathogens and environmental threats.

Scientists fear that the reversed chirality of mirror bacteria could unleash catastrophic consequences: Much like a left-handed nut in a world of right-handed bolts, a mirror bacterium would be incompatible with most immune system molecules, and potential predators would likely fail to recognize it. Although mirror bacteria would not be able to consume natural-chirality nutrients (unless engineered to do so), they would be able to consume available nonchiral nutrients, permitting them to grow. Therefore, scientists worry that mirror bacteria could resist immune defenses, natural predators, and medical and environmental countermeasures, allowing such bacteria to grow with few constraints inside other organisms and the environment and posing an unusual and serious threat to most life on earth (Adamala, Agashe, Binder, et al., 2024).

The Case for Preemptive Governance to Manage the Risks of Mirror Life

Mirror life stands apart from many other emerging risks because of its potential for unprecedented danger to most life on earth and the distinctive structure of this risk, one that limits the usefulness of the strategies on which policymakers typically rely. When facing steep risks from emerging technologies, policymakers often default to two approaches: (1) adaptive governance, in which they try to manage risks as technologies develop by relying on early warning signs and iterative policy adjustment and (2) reactive governance, in which they intervene only after a technology already exists or has begun to spread, typically through restrictive measures, such as export controls or countermeasures.

Such approaches might fail to mitigate the risks of mirror life, however, for the following two primary reasons:

- **There are few reliable early warning signs.** Scientific progress toward a mirror organism depends on breakthroughs in adjacent fields—such as synthetic-cell biology—that advance rapidly and would fall outside mirror life–specific oversight. Work on natural-chirality synthetic cells, for example, is progressing quickly and would substantially lower the technical barriers to creating a mirror cell. Because these enabling advances arise in separate research domains, policymakers might not recognize that mirror life is becoming feasible until the capability already exists. Moreover, incremental steps along the development pathway pose little direct risk on their own, meaning that there are few—if any—signals that would alert policymakers before a viable mirror organism is within reach.
- **Once a mirror organism exists, it may be impossible to prevent its potential harms.** The first successful creation of mirror life would almost immediately lead to diffusion of the know-how required to reproduce or modify it. This knowledge could not reliably be prevented from becoming available to others, including actors who might use it to create more-dangerous strains. Even a nominally safe mirror bacterium could be modified with relative ease into a more harmful version. Once such an organism exists, loss of control could occur through laboratory error or deliberate release, and even a small initial population of mirror life could spread widely before detection and prove impossible to eradicate.

As a result of these challenges in detecting mirror life development early and managing risks once it exists, safeguarding humanity likely will rely on preemptive governance—taking measures before mirror life is created (we explore the need for preemptive governance more thoroughly in Chapter 5). Policymakers can begin developing governance strategies now, before research and development (R&D) activities make the emergence of mirror life a *fait accompli*.

Growing Recognition of the Tremendous Risks of Mirror Life

Discussions about the risks of mirror life are increasing, although formal research into its global security implications remains limited (Adamala, Agashe, Belkaid, et al., 2024; Adamala, Agashe, Binder, et al., 2024). Several convenings have aimed to advance understanding of mirror life. The Mirror Biology Dialogues Fund (MBDF), a philanthropic fund and research advisory organization guided by scientific leaders, brings together scientists, ethicists, security experts, and policymakers to assess aspects of mirror life and its possible governance strategies. In June 2025, MBDF co-organized the 2025 Paris Conference on Risks from Mirror Life with the Pasteur Institute, drawing 335 participants from 31 countries, including the United States, China, Japan, and Germany. The conference issued a statement highlighting that "mirror organisms would present extraordinary immunological and ecological risks while offering limited foreseeable benefits—and that mirror life should therefore not be created" and that "[g]iven these risks, many argued that regulation and law will be needed to ensure that mirror life cannot be created" (Institut Pasteur, 2025a; Institut Pasteur, 2025b). In September 2025, MBDF co-organized an international meeting with the University of Manchester in the United Kingdom (UK) to further discuss the current state of technologies that would enable the creation of mirror life (University of Manchester, 2025).

Additional efforts to assess the risks of mirror life have included a UK science advisory roundtable, which gathered experts to study "the main risks and opportunities associated with researching and developing 'mirror life', including along the pathway of development from cell components to self-replicating cells" (United Kingdom Government Office for Science, 2025). The U.S. National Academy of Sciences also held a workshop in September 2025 to establish a foundation for risk assessment and governance and further shape policy on mirror life, and it is expected to issue a workshop report in 2026.

In addition to these independent risk assessments, several entities have issued statements urging caution. In February 2025, an independently organized meeting to commemorate the 50th anniversary of the 1975 Asilomar meeting on recombinant DNA included sessions on mirror life and led to a statement that was signed by nearly 100 participants from around the world (Abbott et al., 2025). The statement declared: “We believe that mirror life should not be created unless future research convincingly demonstrates that it would not pose severe risks.” In May 2025, a report from the Tony Blair Institute recommended that the United Kingdom also establish an “International Bioengineering Safety Panel to set global research standards and risk controls for high-risk fields like ‘mirror life’ and other forms of synthetic biology” (Macon-Cooney et al., 2025). In July 2025, the United Nations Educational, Scientific and Cultural Organization (UNESCO) International Bioethics Committee released a draft report on the ethical issues related to the research, development, and application of synthetic biology, in which it called for a “precautionary global moratorium on creating mirror cells” (International Bioethics Committee, 2025). Also in July, two signatories of the original warning, along with three additional coauthors, proposed “barriers” or “red lines” beyond which research efforts might stray too close to the creation of a mirror cell (Evans et al., 2025). In September 2025, the German Central Committee on Biological Safety reviewed the arguments made by the authors of the technical report and published a comment on its website stating that it “broadly agrees with their assessments” and that “the demand for a broad scientific and socially orientated debate is explicitly supported” (Zentrale Kommission für die Biologische Sicherheit, 2025). And in February 2026, the U.S.-based Nuclear Threat Initiative and the China Arms Control and Disarmament Association jointly called on the international community to “develop national and international frameworks to guard against the risks posed by mirror life, reaffirming the shared view that such organisms should never be created” (Huaicheng and Yassif, 2026).

RAND has contributed to this emerging dialogue through a 2025 publication, *Policy Options to Prevent the Creation of Mirror Organisms* (Epstein, Crawford, and Nevo, 2025). Others have begun to explore the distinct phases of developing mirror life that can act as barriers to research and provide opportunities for strategic governance and policy options (Evans et al., 2025).

Why the United States Might Need a National Strategy for Mirror Life

In this report, we aim to inform the development of a U.S. strategy to address the risks posed by creation of mirror life. We do not aim to predict whether or when mirror life will be created and do not assert with certainty that its effects would be as catastrophic as Adamala, Agashe, Belkaid, et al. (2024) warn. Rather, **we take seriously the possibility that creation of mirror life could expose most multicellular life on earth to unacceptable risks, and we treat that possibility as sufficient motivation to warrant the immediate development of a strategic plan to better understand the risks and to prevent potential catastrophe**. We intend this strategy to offer U.S. policymakers a road map for preventing the creation of mirror life.

We focus on a U.S. strategy for two reasons. First, U.S. policymakers are our primary audience, and therefore we examine the most constructive role the United States could play in reducing the risks of mirror life. Second, because the United States is both a superpower and a global center of bioscience and biotechnology, its actions will inevitably influence the global trajectory of mirror life development. Steps by the United States could help secure global restraint or, alternatively, spur a global race.

Although what we write in this report is a U.S. strategy, the approach that we outline is inherently global. Preventing mirror life cannot be achieved by any single country, and the United States’ most constructive role is as a partner in cooperation among the major scientific powers, particularly China. What emerges, therefore, is a strategy of shared responsibility: a framework for collective restraint that begins with leadership among those capable of creating mirror life and extends to the broader international community.

We recognize that U.S. leadership in ensuring global restraint is not the only possible organizing principle. Multilateral governance led by a consortium of scientific powers may, for example, be a first step for cooperation within regions, and a United Nations (UN)-based framework might be appropriate for smaller actors. However, given the United States' scientific capacity and international reach, its engagement remains an indispensable component of any effective global prevention strategy, and the feasibility of these other options to create global restraint will depend on U.S. posture.

Importantly, what we present is not intended to be a permanent U.S. strategy; it is a near- to mid-term approach suited to today's world, a world in which the creation of mirror life remains speculative and out of reach because of challenging technical barriers. The strategy's purpose is to maintain these barriers by ensuring that the actors who are most capable of breaching them choose not to.

The world is at a rare inflection point: Mirror life has not yet been created, but the scientific and policy communities are beginning to organize to confront its risks. Mirror life has not yet been politicized, and the topic remains unknown to most publics and governments. This window of opportunity is unusual because factors that could make responsible planning more difficult have not yet emerged. Seizing this opportunity will require a portfolio of governance levers that foster restraint among the scientifically capable actors, make desistance the shared expectation of responsible science, and keep mirror life out of the reach of irresponsible actors who might otherwise pursue it. These levers include incentives and disincentives, regulations, diplomatic tools, verification mechanisms, and norms.

Organization of This Report

We organize this report as follows:

- **Research approach (Chapter 2):** We detail our research approach and its limitations.
- **Actors who might pursue mirror life (Chapter 3):** We catalog the major categories of actors who might pursue mirror life, grouped by shared motivations and capabilities. This actor-level framework provides the basis for assessing the ways in which actors might respond to different levers.
- **Policy levers (Chapter 4):** We organize the broad set of policy levers that could influence various actors into four categories: levers that (1) reduce incentives to engage in harmful behavior, (2) increase barriers, (3) disrupt committed actors, and (4) prepare resilience.
- **Mirror life strategy principles (Chapter 5):** We describe how the distinct scientific and geopolitical features of mirror life create a governance challenge that is distinct from past or existing threats. These distinctions lead to four key principles for the strategy.
- **A U.S. strategy for mirror life (Chapter 6):** Finally, we articulate a U.S. strategy to prevent the creation of mirror life. This strategy assembles the levers (Chapter 4) that can influence actors (Chapter 3); this strategy is consistent with a framework for mirror life (Chapter 5) that ensures that in the end, no actor has both the incentive and the ability to create mirror life.

CHAPTER 2

Research Approach

In this report, we worked inductively from historical analogues, strategic theory, and actor-specific risk pathways to develop a strategy for preventing the creation of mirror life. We first mapped the landscape of actors who might pursue mirror life development and their motivations for doing so, described in Chapter 3. We classified actors into categories and examined their incentives, resources, and constraints, informed by findings from a forthcoming companion RAND report that explores the technical pathways that could lead to creation of mirror life.

We next combined insights from strategic theory, international security studies, behavioral science, examples of historical risk management, and past work by members of our team (Epstein, Crawford, and Nevo, 2025) to create a structured inventory of policy levers that could deter, delay, mitigate, or prevent mirror life development. We evaluated each lever along several dimensions, such as which entities would apply the lever, which actors the lever would affect, whether the lever would affect incentives or capabilities or both, and potential risks and regrets of exercising the lever.

We then developed an underlying foundation for building a U.S. strategy. The unique scientific features of mirror life and the wide variety of actors and motivations for pursuing mirror life together point to several key principles by which levers should be assembled into a strategy. These are presented in Chapter 5. Finally, we bring together the strategic principles, landscape, and policy levers into an integrated U.S. strategy for mirror life in which no actor has both the incentive and the ability to create mirror life.

Assumptions on the Trajectory of Mirror Life Development

The science of mirror biology is nascent and not fully understood even by experts, creating deep uncertainty (Lempert, Popper, and Bankes, 2003) in both scientific knowledge and the effects of policy levers. These uncertainties prevent us from relying on narrow predictions or actor-specific assumptions. They instead motivate the use of robust strategies that remain effective across a wide variety of plausible futures.

We identified the following critical uncertainties that shape the mirror life risk landscape and guide our research approach:

1. whether and how quickly mirror life might be created
2. whether mirror life's creation would cause widespread harm
3. who might create mirror life and why
4. which policy tools could prevent or deter mirror life's creation.

For the first two uncertainties, we assume that mirror life *could* be created and *would* be harmful. We rely on scientists' existing estimates of lead time: between 15 and 30 years at the pace of research prior to their announcement, and possibly within ten years with a "substantial concerted investment" (Adamala, Agashe, Belkaid, et al., 2024; Adamala, Agashe, Binder, et al., 2024). Therefore, **we assume that the creation of mirror life is a credible mid-term possibility that warrants short-term attention**. In this research, we address the remaining two uncertainties, providing answers about who might create mirror life and why and identifying and organizing the policy tools that could deter them.

Key Limitations of This Report

This work has several limitations. First, the analysis presented here relies primarily on conceptual analysis and expert judgment because research on pathways to creation of mirror life and the risk arising from its development is relatively new. We rely on existing assessments by scientists (Adamala, Agashe, Binder, et al., 2024) because independent evaluation of the scientific evidence of the potential for and effects of mirror life is beyond the scope of this report.

Second, we do not address, in detail, the logistical or political challenges that might arise from pursuing elements of the recommended strategic approach. For example, we suggest that strategic decisionmakers consider proposing national legislation that regulates mirror biology research and criminalizes development that comes too close to creation of mirror life. However, the potential difficulties involved in drafting and passing such legislation are outside our scope.

Third, we recommend an approach that we judged to be sound at the time of writing in mid- 2026. The conversation among scientists and policymakers about the potential risks and governance approaches for mirror life is still in its early stages. This rapidly evolving conversation and the parallel efforts to better understand the risks of mirror life could result in changes to the strategic landscape that affect the validity of the strategy we develop in this report.

CHAPTER 3

Actors Who Might Pursue Mirror Life

In this chapter, we catalog the major categories of actors who might be motivated to develop mirror life, examine those actors' underlying goals, and assess those actors' technical and strategic capabilities. For example, although mirror life has no plausible military utility, misperceptions of its strategic advantage could still motivate its pursuit, as could beliefs about its symbolic value. Understanding the range of potential actors is essential for designing effective strategies to either diminish their capacity or dissuade them from pursuing mirror life all together.

To support this goal, we group actors into categories based on shared motivations, resources, and sources of influence. These categories allow us to reason about risks and policy levers at a manageable level of abstraction while still capturing important distinctions among actor types.

The analysis in the chapter reveals an important asymmetry in the mirror life landscape. Although many actors might wish to pursue mirror life for strategic, ideological, or coercive reasons, only a small number of scientifically advanced actors possess the capacity to do so independently. For the foreseeable future, isolated or sanctioned regimes, nonstate groups, and individuals lack the infrastructure, expertise, and access to materials required to create mirror organisms without assistance from the global scientific community.

This asymmetry has an important strategic implication: Because the ability to create mirror life is concentrated among a few scientific powers, if these actors agree that creating mirror life is neither in their individual nor in their collective interest, their restraint effectively denies the means to all others.

Table 3.1 summarizes key actor categories, those actors' technical capacities, their potential motivations, and the factors that may deter them from developing mirror life. The subsequent sections further describe each category of actor. The taxonomy is structured primarily around the approximate magnitude of the resources commanded by the actor. Within categories, actors may differ in their motivations and goals.

Countries with Advanced Biological Science Capabilities

Major nation-states can have gross domestic products (GDPs) in the trillions, substantial bioscience investment, prestigious research universities, and mature pharmaceutical and biomedical industries. The resources required to make serious scientific progress toward mirror life are likely within reach for a wide variety of countries and alliances, including such scientific superpowers as the United States, China, and the European Union, and other countries with advanced scientific research capabilities, including France, Germany, Japan, and possibly Russia.

The United States and China have advanced synthetic biology infrastructure and workforce, large-scale R&D funding, military-industrial and academic integration, and tens or hundreds of trillions of dollars of GDP. Both the United States and China have openly funded academic research on mirror biomolecules and mirror life. Between 2014 and 2024, the U.S. National Science Foundation (NSF) funded research to create a mirror-image biosynthesis system (a first step toward the scientists' long-term goal of creating mirror life) and research that had the explicit goal of creating the first mirror cells (NSF, 2014a; NSF, 2014b; NSF, 2019;

TABLE 3.1

Possible Actors Seeking Mirror Life and Their Capacities, Motivations, and Barriers

Actor Category	Actor(s)	Scientific Capacity to Create Mirror Life	Financial Resources Available	Potential Motivations for Developing Mirror Life	Potential Barriers to Developing Mirror Life
Countries with advanced biological science capabilities	Includes the United States, China, Germany, the United Kingdom, France, Russia, and Japan	• World-leading capabilities • Advanced synthetic biology capabilities • Large R&D sector • Skilled workforce	• GDP in the trillions, advanced science research infrastructure	• Achieve economic, military, or scientific leadership • Prestige • Scientific advancement	• Democratic constraints • Risk aversion • Perception of potential global harm • Lack of military utility or clear commercial benefit • International norms and example-setting
Isolated states and sanctioned regimes	Includes North Korea and Iran	• Limited • No well-funded bioscience sector • Can redirect resources • May seek foreign help or theft	• GDP in the tens or hundreds of billions, strong military	• Regime survival, fueled by deep suspicions of subversion by the West • Perceived value in asymmetric deterrence, bargaining • Destabilization of adversaries	• Technical barriers • Lack of expertise and resources • Credibility issues • Risk of self-destruction
Private commercial enterprises	Multinational biotechnology firms, pharmaceutical firms, biotechnology startups	• Varies • Some have advanced R&D • Some can attract top talent • Less basic science capability than academia	• Varies; large firms have revenues comparable with small countries	• Profit, innovation, commercial applications (e.g., therapies, biomanufacturing) • Prestige	• Regulatory and legal barriers • Reputational risk • Lack of consensus on safety • Market disincentives
Academic and private research institutions	Includes major top-tier universities in the United States, China, and other countries	• High capabilities in basic science • Strong human capital	• Varies; some institutions have billion-dollar endowments (with restrictions on research use), mostly rely on grants and external funding, including venture capital funds	• Scientific curiosity and achievement • Prestige • Patenting and eventual commercialization	• Ethical, biosafety, and biosecurity concerns • Institutional oversight • Funding limitations • Public accountability
Transnational criminal and violent extremist organizations	Includes the Islamic State of Iraq and Syria (ISIS), al Qaeda, and drug cartels	• Low • Rarely have advanced technical expertise • May recruit scientists	• Varies; some have resources as large as small countries, but have limited resources	• Coercion and profit (criminals) • Ideological goals (extremists) • Perception of deterrence or bargaining power	• Lack of technical expertise • Difficulty recruiting scientists • High risk of detection

Table 3.1—Continued

Actor Category	Actor(s)	Scientific Capacity to Create Mirror Life	Financial Resources Available	Potential Motivations for Developing Mirror Life	Potential Barriers to Developing Mirror Life
Apocalyptic fringe groups	Includes successors to Aum Shinrikyo, other apocalyptic cults	• Very low • Little technical capacity • Small size	• Very limited	• Apocalyptic or nihilistic goals • Insensitivity to deterrence	• Extreme technical barriers • Lack of resources, small size, inability to develop independently
Well-resourced lone individuals	Ideologically motivated individuals, disgruntled scientists	• Minimal • May have access via employment or theft • Not independent creators	• Ranges—some individuals have more resources than small countries	• Personal ideology • Psychopathology, desire for notoriety or harm • Insensitivity to deterrence	• Cannot develop independently

Weidmann et al., 2019). In China, government-funded work related to mirror life has focused on creating biological systems that are capable of synthesizing mirror DNA, RNA, and proteins, which are prerequisites for the creation of mirror life (Fan, Deng, and Zhu, 2021; Jiang et al., 2017; Ling et al., 2020; Wang et al., 2016; Xu et al., 2017; Xu and Zhu, 2022). Many U.S. and Chinese scientists who have previously conducted research related to mirror biology have called for caution and governance of research that could enable the creation of mirror life (Abbott et al., 2025; Adamala, Agashe, Belkaid, et al., 2024; Evans et al., 2025; Zhu, 2025).

Aside from direct work on mirror biomolecules and mirror cell components, both the United States and China have funded a wide variety of basic science and synthetic biology research and technology development that could enable further advancements toward mirror life.

Strategic decisionmaking in both China and the United States is informed by research and risk assessment. Both countries possess powerful conventional and nuclear weapon arsenals, neither seeks global annihilation, and neither would be likely to pursue mirror life as a biological weapon for the reasons that we describe in Chapter 4. Restraint from developing mirror life could change if either country doubts the scientific evidence for its risks or perceives potential economic, commercial, or deterrence benefits and believes that mirror life could be pursued with acceptable risk. However, as with any emerging technology, perceptions of risk and benefit can evolve. If either country were to misjudge the science or fear being left behind, that restraint could erode. Continued transparency and international engagement are essential to prevent such missteps.

Several medium-sized powers also have advanced biotechnology capabilities. Germany has openly funded scientific work to establish a self-replicating mirror-image biological system (Adamala, Agashe, Binder, et al., 2024; Weidmann et al., 2019). But other countries are gaining awareness of possible risks from mirror life and considering governance options. For example, a UK expert roundtable (United Kingdom Government Office for Science, 2025) studied the possible benefits of mirror life, the risks of developing mirror cells, and potential mitigations of such risks.

Medium-sized powers account for the majority of the world's nine confirmed nuclear weapon states. With the notable exception of Russia, most of these countries have traditionally fielded a so-called minimum deterrent that is intended to inflict unacceptable damage on an aggressor rather than to prevail in an all-out nuclear conflict. It is conceivable that a medium-sized power might try to exploit the threat of unleashing mirror life as either an addition to or a substitute for nuclear deterrence. Again, however, rational leaders of these states likely would also find mirror life to be too dangerous a tool for deterrence and would be expected to desist.

Russia is a special case in this category. It boasts greater military power than one would expect for an economy of its size. And although Russia's biological research establishment is not what it was during the Cold War, it is still formidable enough for Russia to be a viable competitor if it were to prioritize state resources for advancing mirror life. Moscow's pursuit of biological weapons during the Soviet period and the contemporary allegations of continued biological weapons research make it particularly important to understand the Russian government's intentions around mirror life. However, again, rational decisionmakers who believe scientists' warnings about the potential dangers of mirror life would not be expected to develop or deploy a weapon that they understood would destroy themselves and the rest of the world.

Isolated States and Sanctioned Regimes

States that are politically isolated or under sanctions, such as North Korea and Iran, have historically been central to concerns about weapons proliferation, particularly where regimes prioritize strategic capabilities over domestic welfare. However, most such states lack the scientific infrastructure and funding, as well as the human capital, that are required to pursue highly advanced synthetic biology programs. Accordingly, these states are unlikely to develop mirror life on their own under existing conditions. The more plausible concern is that scientific advances from other high-resource countries could lower the technical barriers over time, making it possible for states with limited scientific abilities to develop mirror life in the future.

Once this happens, these states could be motivated to pursue mirror life as a bargaining chip to be traded for other regime priorities (such as sanctions relief). However, a state that sought to use the threat of mirror life this way would face a credibility problem: A state that values its own survival and the future benefits that it seeks would be unlikely to deliberately trigger global annihilation if its demands were not met.

Alternatively, some states might also seek mirror life as a doomsday capability, hoping that the threat of global annihilation would deter other countries from acting against them. Yet even this logic presents difficulties. If deployment were discretionary, the threat would again lack credibility; if deployment were automatic following an attack—analogous to the doomsday machine logic articulated by Herman Kahn in the context of nuclear deterrence (Kahn, 1965)—the regime would be committing itself to its own destruction. Moreover, such a capability or evidence of its pursuit might incentivize extreme international preemption or internal resistance. Importantly, however, history demonstrates that some regimes are willing to accept self-destruction in pursuit of ideological, religious, or existential goals.

As with nuclear weapons, such strategic paradoxes may not prevent pursuit, but they do substantially constrain the plausibility of mirror life as a usable or stable instrument of state strategy. Moreover, the perception that mirror life could have strategic value—even if misguided—could incentivize some states to pursue it once it is within their technological reach.

Private Commercial Enterprises

Private commercial enterprises vary enormously in the resources available to them. The largest multinational firms boast revenues greater than the GDPs of small countries, and biotechnology startups sometimes raise substantial capital to pursue promising R&D projects. These entities might seek to create mirror life in pursuit of commercial innovation or prestige if they believe that mirror life would not be capable of catastrophic harm. Given that private enterprises exist to make a profit, these firms would most likely seek to create mirror life in pursuit of its commercial potential if they judged that its possible benefits outweigh its risks and costs.

Posited commercial or beneficial applications of mirror organisms are speculative but could include developing novel cell therapies, improving biomanufacturing because the cells could resist outside infection, and using mirror organisms to produce mirror biomolecules more cheaply and accurately than existing methods (Adamala, Agashe, Binder, et al., 2024).

Commercial enterprises might not recognize or prioritize the biosecurity and biosafety risks of mirror life development. However, if some of these steps or products were regulated or banned, this would effectively dissuade most commercial actors from pursuing the development of mirror life. Although some startup companies seeking “disruptive” biotechnologies might be less responsive to regulatory or legal risk, most private commercial enterprises operate within market and regulatory structures and value their firms’ reputations. If consensus developed among the international community that mirror life development posed world-threatening risks, or if pursuit of mirror life constituted a crime in important business jurisdictions, it does not seem likely that any legitimate enterprise that wished to retain its legitimacy would pursue the creation of mirror life, even if it could find an environment in which such research remained legal.

Academic and Private Research Institutions

Synthetic biology laboratories within academic research institutions could pursue mirror life for a variety of reasons: curiosity, achievement, prestige, or as a potential spinoff for eventual commercialization. Universities could also fund or support research that would enable or create mirror life and would be aligned with funding priorities, even if national funding for such research were to be withheld or regulated. Academic researchers could inadvertently advance mirror biology by pursuing enabling technologies for goals other than the creation of mirror life.

Most research conducted within academic institutions is released publicly through scholarly publications and other dissemination outlets. In fact, many universities have well-funded media relations offices that contribute to the public dissemination of scientific findings. Most institutions also have well-developed institutional biosafety and biosecurity policies and perform ethics reviews for research that might create risks.

The resources available to academic researchers vary enormously. Although a few universities have billion-dollar endowments, much of that funding is typically restricted and would not be available for research (Ghaedi, 2025). Moreover, most schools are comparatively dependent on tuition and government research grants for their operations (Ghaedi, 2025). Even the wealthiest schools usually draw on outside sources of funding to support biomedical research. For example, the most prestigious U.S. universities mostly rely on funding from the NSF and the National Institutes of Health to support their research (Nietzel, 2024). This reliance can result in academic researcher priorities that reflect national interests.

Those working in academia pose little risk of having malicious intent, but there is significant risk that they can advance foundational science because of academia's vast human capital and technical capability. Despite the temptation of more pay in the private sector or government labs, many of the world's most talented biology researchers choose to make a career in academia. Consequently, academic researchers have a strong capacity for basic science, including mirror-image biochemistry.

Transnational Criminal and Violent Extremist Organizations

Transnational criminal organizations and violent extremist organizations vary enormously in both their motivations and resources. The key difference between transnational criminal organizations and violent extremist organizations is that the former are primarily motivated by profit, while the latter pursue ideological ends. However, in practice, the distinction between these two categories is not always clear-cut (Byman, 2015). For example, some transnational criminal organizations are set up by ideologically motivated groups as a means of fundraising. Although the motivations of these groups might be political, territorial, or financial, they do not, in general, seek annihilation of all life on earth. In this way, these organizations are similar to isolated or sanctioned states but have fewer institutional and financial resources. If technology or knowledge helpful for creating mirror life were widespread, these actors might seek to create it.

A key weakness of these types of groups is that they usually have difficulty recruiting scientific and technical experts. This is not universally true—drug cartels manage to train or recruit chemists to enable their production of illicit drugs (Kitroeff and Villegas, 2024)—but leading-edge technically talented individuals usually eschew the lifestyle of the criminal and the insurgent. The creation of mirror life would also constitute a scientific achievement far beyond the chemical synthesis of illicit drugs. These groups would have to rely on recruitment to fill their ranks with scientific experts, and they are unlikely to have technical R&D programs on par with even a small nation-state. Criminal or terrorist organizations could conceivably pursue mirror life cooperation with a nation-state or through covert partnerships with scientists with access to the necessary resources.

Apocalyptic Fringe Groups

Although so-called doomsday cults are uncommon, they do, in fact, exist and are one of the most worrisome constituencies of concern for mirror life risk. In the 1990s, the Japanese apocalyptic cult Aum Shinrikyo carried out an attack with nerve gas and pursued other weapons of mass destruction (WMDs), including nuclear weapons (Thornton, 2018). Members of these groups may have extreme, apocalyptic beliefs that differ so radically from typical values that conventional deterrents, such as imprisonment or the threat of death, may have little effect.

Although ideologically motivated fringe groups might desire mirror life for their nefarious ends, the extreme technical complexity of developing mirror life makes such efforts implausible in the near future. Apocalyptic extremist groups are typically small and command modest resources, even by the relatively cash-strapped standards of terrorist organizations. But if the means to create mirror life were sufficiently developed and accessible to nonstate actors, these groups might be able to leverage the work of others to acquire mirror life. If the technology to create mirror life becomes sufficiently diffused, the probability of apocalyptic cults acquiring such technology could become uncomfortably high.

Well-Resourced Lone Individuals and Insider Threats

A small number of independently wealthy or technically sophisticated individuals could seek to create or possess mirror life; two such reasons would be notoriety-seeking or severe psychopathology. Such actors would probably be insensitive to conventional deterrent threats. However, even well-resourced individuals are unlikely to possess the full scientific infrastructure and knowledge required to independently create mirror life. Their primary risk pathway would therefore involve attempting to finance research and contract expertise, rather than developing the capability entirely on their own.

Relatedly, insider threats arise from individuals embedded within legitimate research institutions. These actors need not be wealthy or independently capable. The risk that they pose stems from authorized access to resources, such as facilities, materials, or sensitive knowledge. An insider could intentionally divert materials, smuggle samples, or misuse ongoing research conducted by well-resourced organizations. If mirror life–enabling research is permitted, institutions conducting such work would need programs and monitoring mechanisms, as well as access controls, to mitigate insider diversion risks.

CHAPTER 4

Potential Levers to Address the Development of Mirror Life

In this chapter, we catalog the many levers in the U.S. governance toolbox from which we assemble a U.S. strategy for mirror life (presented in Chapter 6). We include those policy levers with historical precedent that might be feasible for the United States and other concerned countries to use. Some of these levers might require substantial political or financial expenditures. For completeness and transparency, we also include levers that might be problematic because they pose serious risks, are incompatible with other levers, or may prove to be ineffective given the specific nature of mirror life. In Chapter 6, we present our assessment about whether and how such levers should be included in a U.S. strategy.

To help readers make sense of the large catalog of levers, we organize levers as shown in Table 4.1 using a structure adapted from widely applied prevention frameworks in public health, criminology, and counterproliferation. Public health distinguishes between primary, secondary, and tertiary prevention—reducing underlying drivers of harm, detecting and interrupting early stages, and mitigating consequences when prevention fails (Gordis, 2013; Leavell and Clark, 1965). Criminology's situational prevention framework similarly emphasizes reducing incentives, increasing the level of effort required to engage in harmful activity, and raising the risks of detection (Clarke, 1995). Modern counterproliferation and biodefense strategies commonly employ a prevent-detect-interdict-respond sequence to reduce the likelihood and impact of dangerous technological developments (National Academies of Sciences, Engineering, and Medicine, 2018; U.S. National Security Council, 2018). Although the traditions in these three fields differ, they share a common logic: Reduce incentives to engage in harmful behavior, increase barriers, disrupt committed actors, and prepare resilience.

We apply this prevention logic to the mirror life context to create the following categorization of policy levers:

1. **Reduce incentives to develop mirror life.** These levers make mirror life unnecessary or unattractive by advancing scientific knowledge of its risks, supporting the development of safer alternatives, strengthening public and ethical norms against its creation, and securing agreements in which states or institutions commit not to pursue mirror life research.
2. **Detect and deter early-stage development.** These levers (1) improve the ability to know whether mirror life research is occurring and (2) make early development risky to attempt. Such actions include strengthening detection, monitoring, and verification capabilities and raising the costs of initiating such research through funding restrictions, liability mechanisms, regulatory controls, and criminalization.
3. **Disrupt or punish advanced development.** These levers prevent progress once an actor is already committed to developing mirror life by interfering directly with their ability to continue. The relevant tools include imposing sanctions, restricting visas or academic access for involved individuals or institutions, and disrupting supply chains or cyber infrastructure. However, as we discuss in

Chapters 5 and 6, attempts to apply these tools internationally carry significant risks of escalation, collateral harm, and unintended signaling. Thus, the use of these tools is considered counterproductive and outside the scope of our proposed U.S. strategy, which, as explained in the following chapters, focuses on prevention.

4. **Build countermeasures and resilience.** These levers aim to limit harm if mirror life is developed, including with medical and environmental countermeasures and measures to bolster societal resilience under extreme conditions. We argue that countermeasures would be largely ineffective and that their premature development could introduce new risks by signaling acceptance of mirror life or triggering destabilizing dynamics. Accordingly, countermeasure and resilience efforts should be considered only under carefully managed, transparent circumstances. We discuss these roles and limitations in greater detail in Chapter 6.

Within each category, we describe the levers, the actors they would influence and how, what other levers are enablers or enabled, and the risks of such levers, particularly amid uncertainty about the risks of mirror life itself. The placement of individual levers within these four categories is not always clear-cut because many tools operate through multiple mechanisms. A global scientific oversight committee, for example, could primarily shape norms and reduce incentives to pursue mirror life while also serving as a monitoring body that supports early detection. Likewise, levers that block, disrupt, or punish advanced development can have second-order deterrent effects on early-stage actors. In this framework, we place each lever where it best fits the dominant logic of its operation while acknowledging that reasonable alternative categorizations exist.

Table 4.1 summarizes the levers described in this chapter.

Levers to Reduce Incentives to Develop Mirror Life

A strategy to prevent the creation of mirror life may aim to reduce incentives for its development altogether. This section outlines a variety of policy levers—such as targeted risk assessments, support for safer alternatives, coordinated public messaging, establishment of ethical standards, and international legal and diplomatic agreements—that can help prevent mirror life from becoming an attractive or viable pursuit for any actor.

TABLE 4.1

Potential Levers to Address the Threat from Mirror Life

Lever Category	Levers
Reduce incentives to develop mirror life	• Conduct additional research to address remaining uncertainties • Support the development of safer alternatives to mirror life • Issue public and multilateral statements and resolutions • Develop voluntary and ethical standards • Establish agreements not to pursue mirror life
Detect and deter early-stage development	• Expand monitoring • Develop verification methods • Establish a global scientific oversight committee • Increase direct economic barriers • Restrict access to enabling materials, technologies, and knowledge • Criminalize and prosecute domestic mirror life development
Disrupt or punish advanced development	• Impose sanctions and economic isolation • Apply visa and academic restrictions • Conduct operational disruption
Deploy countermeasures and build resilience	• Develop medical and environmental countermeasures • Enhance societal resilience

Conduct Additional Research to Address Remaining Uncertainties

Public and philanthropic funders can play a central role in advancing scientific understanding of mirror life's risks. This includes research on health and ecological hazards, detection methods for incremental progress, and potential chokepoints—scientific or technical milestones that, once surpassed, could sharply accelerate progress toward the first mirror organism.

Research to reduce uncertainties can be conducted safely without advancing the creation of mirror life itself and is likely necessary to justify stronger policy action. Although some risk exists that publicly funded research into mirror life hazards could inadvertently broaden access to enabling knowledge, actors intent on developing mirror organisms would likely pursue such knowledge regardless. In contrast, rigorous research on risks and vulnerabilities can help motivate and legitimize restrictions aimed at preventing those same actors from proceeding while minimizing the impact of such restrictions on research that is unrelated to mirror organisms and that does not provide actionable steps toward creating mirror life.

In the United States, such agencies as the NSF, National Institutes of Health, and major research foundations could support such research. The National Academies of Sciences, Engineering, and Medicine can also contribute through independent, consensus-based studies commissioned by either governmental or nongovernmental funders. These assessments can clarify remaining uncertainties around the risks of mirror life, strengthen the evidence base for those risks, and provide policymakers with trusted guidance on emerging technological risks.

Additional research could validate emerging scientific consensus that the risks of mirror life outweigh any plausible benefits, especially if that research incorporates expertise from diverse disciplines and countries. Research should focus on key uncertainties—for example, the ecological processes most vulnerable to disruption, the earliest detectable signals of accidental or deliberate release, the potential consequences of research restrictions, the viability of technical alternatives to mirror life, and the feasibility of policy alternatives. The U.S. experience phasing out chlorofluorocarbons (CFCs), driven by coordinated scientific research and policy action following the 1977 Clean Air Act amendments, illustrates how early assessment can lay the foundation for effective governance of emerging environmental threats (Manzer, 1990).

Support Development of Safer Alternatives

Funders can also support research on safer alternatives to mirror life. Alternatives include, for example, methods for producing mirror biomolecules without creating self-replicating mirror organisms or technologies that would materially advance mirror life development. These approaches can lower incentives for commercial or academic actors who might otherwise pursue mirror life for biomedical, biomanufacturing, or scientific applications.

At the strategic level, alternatives also help address a central challenge for global restraint: concerns that forgoing mirror life could mean surrendering meaningful opportunities. Alternatives reduce the perceived opportunity cost of restraint by demonstrating that the conceivable (if unproven) benefits of mirror life can be achieved through safer means. This makes collective restraint far more politically and scientifically sustainable, especially for commercial actors and scientifically advanced states that might otherwise fear being left behind.

There is some risk that improved alternatives could indirectly support mirror life research. However, reducing the perceived scientific or commercial value of mirror organisms by providing safer substitutes might be a stronger deterrent than restricting alternatives. Research into alternatives therefore complements risk assessment by reducing the appeal of mirror life itself while avoiding work that would materially advance its creation.

Issue Bilateral and Multilateral Statements and Resolutions

Coordinated public messaging that is delivered by national governments, scientific institutions, international bodies, and respected experts can delegitimize mirror life development before the idea gains technical plausibility or social license. Such messaging should emphasize that no scientifically supported benefits outweigh the risks of creating a self-replicating mirror organism and that even the strictest biosafety systems cannot guarantee perfect containment.

This raises reputational costs for universities, companies, and governments that are considering mirror life research. It might also indirectly deter extremist groups or fringe actors by reducing the ideological appeal of mirror life and stigmatizing associated research.

A particularly salient example is laboratory containment risk. Even biosafety level (BSL)-4 laboratories, designed for the world's most dangerous pathogens, have had leaks in the past (Wurtz et al., 2016). Although the accidental release of a conventional pathogen could cause severe illness or even a pandemic, humanity can recover from such an event. Accidental release of a mirror organism, by contrast, could irreversibly threaten global ecosystems and the viability of most forms of life on earth. Clear messaging that draws on rigorous scientific risk assessments can frame the pursuit of mirror life as reckless, unethical, and scientifically unjustifiable.

Public messaging plays two roles. First, it serves as an early, low-cost disincentive: Statements require no new legislation or treaty infrastructure and can be issued immediately by federal institutions (e.g., the U.S. State Department, the White House Office of Science and Technology Policy). Second, public dissuasion that engages diverse stakeholders can cultivate political momentum and legitimacy for later binding actions, such as legislation, multilateral agreements, export controls, and enforcement tools.

Options for public and multilateral messaging include the following:

- **National statements** from U.S. agencies or scientific bodies declaring that the risks of mirror life vastly exceed any hypothetical benefits and that the United States will not pursue such research. Such messaging can echo historical precedents, such as the U.S. renunciation of biological weapons.
- **International or multilateral statements** from forums such as the UN General Assembly, UNESCO, or the World Health Organization (WHO). As precedent, the January 1992 UN Security Council statement characterizing WMD proliferation as a threat to international peace and security shows how political norms can be set even without binding law (UN Security Council, 1992).
- **Moratoria declared by international scientific bodies**, similar to proposals on human cloning or AI pause efforts (e.g., Future of Life Institute, 2023; Mayor, 2001), which shape expectations even if not broadly adopted.
- **Narrative strategies** to undermine any ideological, philosophical, or symbolic appeal of mirror life. Drawing from post-ISIS counter-radicalization efforts, coordinated messaging could portray the pursuit of mirror life as irresponsible, dangerous, unscientific, or fundamentally contrary to global ecological stewardship.

The pitfalls of public messaging are modest. Public messaging can provoke backlash if it appears moralizing or motivated by geopolitical advantage. Actors skeptical of U.S. intentions may interpret U.S. messaging as an attempt to constrain competitors or monopolize biotechnology leadership. To mitigate these risks, messaging should be evidence-based, transparent about uncertainties, and ideally coordinated with other scientific powers—especially China—to show shared scientific concern rather than unilateral political positioning. Additionally, discussing catastrophic risks could inadvertently attract attention from omnicidal actors; however, withholding credible analysis would hinder responsible actors far more than it would reduce awareness among malign ones.

Develop Voluntary and Ethical Standards

Voluntary standards, professional guidelines, and expanded institutional review can define responsible scientific conduct and categorize mirror life–enabling research as ethically unacceptable. Although these mechanisms are nonbinding, they disincentivize mirror life development by increasing reputational and procedural barriers for researchers, universities, private-sector developers, and funders seeking to conduct high-risk work. They also create early-warning visibility, helping funders and governments monitor areas in which research is converging toward mirror life–relevant capabilities.

Historical examples demonstrate the effectiveness of voluntary or quasivoluntary scientific governance. The Asilomar Conference led to the creation of institutional biosafety committees and funding restrictions on recombinant DNA research long before comprehensive regulation existed (Berg, 2004). Today, AI ethics frameworks—such as Institute of Electrical and Electronics Engineers (IEEE) global standards—show how voluntary norms can inform expectations across diverse technical communities. A similar approach in mirror life governance could stigmatize enabling research and align scientific institutions with global restraint.

Enhanced oversight could build on existing mechanisms for high-consequence biological research. Since 2012, U.S. life-science research policies have required special review of certain dual-use or high-risk projects (Office of Science and Technology Policy, 2014). The Federal Select Agent Program applies oversight to all work that involves designated pathogens, regardless of funding source. A comparable oversight regime could require registration or review for research that materially lowers barriers to creating self-replicating mirror systems. Because such a regime would apply beyond federally funded environments, new legislation would likely be needed.

Options for voluntary or ethical governance include

- **voluntary standards** developed by professional groups or international bodies that categorically reject mirror life development and require heightened scrutiny for mirror-adjacent research.\
- **expanded institutional review**, including institutional review board–like or biosafety committee review for any work that significantly reduces technical barriers to mirror life creation
- **registration and external oversight** for research programs whose tools, knowledge, or methods meaningfully facilitate mirror life development, including a requirement for institutions to certify that their projects do not cross red lines
- **global professional ethics guidance**, issued by scientific societies or academies, signaling that mirror life research violates professional responsibility and endangers global biosphere stability.

Developing voluntary standards requires care. Standards might be dismissed as symbolic if they lack clear scientific justification or if they appear to be unevenly enforced across countries. Excessively restrictive norms risk pushing research offshore to jurisdictions with weaker oversight, creating fragmentation rather than restraint. Conversely, overly permissive standards could inadvertently legitimize dangerous research. Maintaining credibility requires standards to evolve with scientific understanding and to be coordinated among major scientific powers so that norms are not interpreted as asymmetric constraints designed to disadvantage competitors.

Establish Agreements Not to Pursue Mirror Life

A comprehensive approach to dissuading the development of mirror life must combine technical constraints with legal and diplomatic pressure. Multiple diplomatic and legal pathways are available to reinforce global restraint. These tools work by raising the legal and strategic costs of pursuing mirror life activities.

Some actors may believe that catastrophic risks are overstated or that mirror life could be exploited for strategic advantage. These actors may be influenced by security guarantees or confidence-building mechanisms—

for example, expanded defense partnerships or mutual restraint agreements. Such measures can reduce the pressure on states to seek destabilizing technologies by providing alternative sources of security.

Diplomatic and legal efforts would primarily be led by foreign ministries, treaty organizations, UN bodies, and strategic alliances. These efforts' purpose would be to establish global legal norms, create diplomatic consequences for transgression, and reassure states that others are refraining as well. These tools would primarily shape state behavior but also would indirectly affect private-sector and research institutions that align their actions with prevailing global norms.

Several pathways exist for building these commitments:

- **International resolutions:** International resolutions—such as those issued by the UN General Assembly or specialized UN agencies—offer early, low-cost signals of global concern. They can frame mirror life development as being outside the realm of responsible state behavior, emphasize the absence of a legitimate peaceful purpose for mirror life development, and encourage states to align their domestic policies on such development accordingly. These resolutions also prepare the diplomatic landscape for more-formal agreements by normalizing expectations that mirror life development is globally unacceptable.
- **Existing treaties or institutions:** Existing international legal regimes may offer avenues for incorporating mirror life into binding obligations. States parties to the Biological Weapons Convention (BWC) could assert that mirror organisms fall under treaty prohibitions because they lack peaceful utility. The UN Security Council could adopt a Chapter VII resolution mandating restrictions, drawing on such precedents as UN Security Council Resolution 1540 (UN Security Council, 2004). Other treaties—such as the Environmental Modification Convention (ENMOD)—could be invoked if mirror life is framed as an environmental threat. Even if they are politically difficult to accomplish, these pathways reinforce the idea that mirror life development has shared security implications and warrants binding constraints.
- **Bilateral (e.g., U.S.-China) diplomacy and confidence-building:** Bilateral agreements can reduce the incentives for unilateral development and reassure states of mutual restraint. Examples include reciprocal no-development assurances, shared early-warning mechanisms, or coordinated export controls. Historical arms-control examples—such as Cold War U.S.–Soviet agreements or U.S.–European Union technology-transfer coordination—demonstrate the potential for bilateral diplomacy to limit proliferation and shape strategic stability, although nonproliferation agreements have not always been successful.
- **Nonproliferation and export-control groups:** International groups, such as the Australia Group, the Proliferation Security Initiative (PSI), or the North Atlantic Treaty Organization (NATO), can harmonize export controls and coordinate supply-chain vigilance, as well as share information on illicit procurement. These forums can meaningfully restrict access to enabling materials or technologies. However, because they exclude major scientific actors, such as China and Russia, their roles must be calibrated to avoid reinforcing perceptions that mirror life governance is a geopolitical project rather than a global risk initiative.
- **New treaty on mirror life:** The most ambitious option is negotiating a standalone treaty that bans mirror life development. Such a treaty could define prohibited activities, mandate domestic legislation, establish verification systems, and create compliance mechanisms. Precedents include the Chemical Weapons Convention, BWC, Montreal Protocol, and the Outer Space Treaty's Article IX (Organisation for the Prohibition of Chemical Weapons, 1997; UN Environment Programme, 1989; UN General Assembly, 1966). However, this pathway requires extraordinary political will and would likely involve lengthy, resource-intensive negotiations. At the same time, a mirror life treaty likely would not share many of the concerns (e.g., intellectual property [IP], equity) that have plagued other recent treaty negotiations. Treaty negotiations might not be the best vehicle for arriving at international consensus on

banning mirror life development, but a treaty could nevertheless be one of the best ways of codifying such a consensus once agreed to.

The effectiveness of diplomatic and legal actions depends on several enabling conditions: shared scientific assessments, transparent engagement among major powers, and visible domestic restraint. Without these conditions, agreements might stall or yield fragile outcomes. Not every venue should be engaged simultaneously; some forums may be redundant or counterproductive, making careful selection and sequencing essential. Domestic political dynamics further complicate implementation. Resistance to such measures as funding limits, export controls, or new regulatory requirements could slow progress and weaken the credibility of U.S. commitments abroad. Relations with competitors also shape outcomes. If adversaries believe the United States is secretly preserving its own advantage, even earnest diplomatic initiatives might appear self-serving. Transparent engagement and demonstrable restraint may help mitigate this risk.

Multilateral diplomacy is also constrained by the broader geopolitical environment. Tensions among major powers often spill into international forums, such as the BWC, UN Security Council, or UN General Assembly, making consensus difficult even among nominal partners. Verification challenges introduce additional fragility: Weak oversight or uneven implementation can render agreements ineffective. Diplomatic fragmentation poses another risk; parallel engagement across the UN General Assembly, UN Security Council, BWC, ENMOD, Australia Group, and PSI can produce conflicting expectations or alienate key actors whose participation is essential for global restraint. Failed diplomatic efforts can be worse than none at all because unsuccessful attempts to secure resolutions or treaty language might be interpreted as evidence that concerns about mirror life are overstated, emboldening would-be developers. These risks highlight the importance of grounding diplomatic and legal levers in early scientific collaboration, visible domestic restraint, and sustained transparency.

Levers to Detect and Deter Early-Stage Development

A credible prevention strategy requires tools that both identify early movement toward mirror life development and discourage actors from beginning to move along that trajectory at all. Early-stage activities—from synthetic cell construction and ribosome engineering to improvements in mirror protein synthesis—may have legitimate scientific value but can also lower technical barriers to creating mirror organisms. Without mechanisms to detect such activity and raise the costs of pursuing enabling research, governance efforts risk being unenforceable or easy to evade.

Accordingly, in this section, we examine levers that (1) expand monitoring and verification, strengthening the ability to detect concerning activity early, and (2) deter initiation of such work by creating barriers that shift incentives away from mirror life development.

Each of these levers carries not only potential benefits, however, but also risks of backlash or counterproductive responses if poorly designed or implemented unilaterally. Measures intended to deter or detect early-stage mirror life research may encourage concealment or provoke geopolitical friction, particularly in low-trust environments or when legitimate scientific activity is affected. For this reason, the effectiveness of each lever depends not only on its technical merits but also on careful design and coordination—considerations that are addressed in the subsections that follow.

Expand Monitoring

Governments can develop and refine collaborative monitoring efforts to identify research programs that could enable mirror life, especially privately funded or high-risk projects. Existing monitoring approaches used for nuclear and biological weapon programs—such as open-source intelligence, satellite surveillance, supply-chain tracking, and confidence-building measures—offer precedents (Bertherat et al., 2024; BWC, 1975; Richelson, 2006). When applied to mirror life, these monitoring approaches could help identify signatures of enabling research and support attribution when breaches occur. Importantly, these tools would focus on identifying concerning research activity, not on treating mirror life as a traditional weapon threat.

In the United States, enhanced monitoring could include adjusting priorities within the Bureau of Arms Control and Nonproliferation at the U.S. Department of State to foster discussions about work on mirror biology with international partners and develop indicators and warnings. Increased analytic focus on enabling research would give policymakers earlier situational awareness and decisionmaking space if prohibited programs emerge.

Monitoring efforts face significant technical and political constraints. Because mirror life–enabling research may resemble legitimate synthetic biology, monitoring programs may generate false positives or miss covert actors entirely. Monitoring could also be perceived as politically motivated or discriminatory, eroding trust among scientific powers and provoking hedging behavior. Some states might respond to such efforts by improving concealment rather than desisting. Finally, uneven monitoring capacity across countries could create global blind spots that undermine deterrence.

Develop Verification Methods

Verification is essential for enforcing restrictions on research that could create mirror life. Robust verification capabilities support domestic enforcement and enable trust-building and transparency among countries.

Effective verification will require having tools that can detect specific and complex mirror biological molecules or systems or other indicators that research has progressed beyond acceptable limits. Although methods could build on biological forensics and arms-control verification, new scientific investment will likely be required to develop reliable techniques for identifying mirror materials.

Verification is most effective when implemented early, before research progresses to stages that could enable self-replicating mirror organisms. Yet verifying bona fide mirror life (e.g., distinguishing self-replicating mirror cells from nonliving or nonreplicating mirror biomolecules) would be extremely difficult without risking accidental release. This underscores the importance of detecting enabling research far upstream.

Historical regimes show both the potential and limits of verification. The Chemical Weapons Convention demonstrates that facility declarations and the threat of onsite inspections may deter major violations, but its focus on large-scale production leaves smaller activities undetected (Boehme, 2008). The BWC offers a cautionary contrast: Lacking a verification regime, it has struggled to detect illicit biological activity, partly because of the small scale of biological facilities and their dual-use nature. Mirror life poses an even greater challenge, suggesting that verification cannot simply copy existing arms-control models.

Verification for mirror life will be inherently constrained by scientific uncertainty and by the difficulty of detecting small, concealable research. Intrusive inspections might face political resistance, and ambiguous findings could fuel disputes or accusations of bias. Overreliance on early-stage or immature verification tools may create false confidence, while adversaries may exploit gaps to conduct covert development. Imported weaknesses from past verification regimes could undermine credibility if they are not addressed.

Establish a Global Scientific Oversight Committee

A global scientific oversight committee could provide a central forum for international transparency and governance. The committee could review emerging risk assessments, evaluate research trends, advise on safe practice standards, and help distinguish acceptable scientific work from activity that meaningfully lowers barriers to mirror life. This oversight body could be housed within a UN agency, formed through an international scientific consortium, or constituted as an independent entity.

A useful analogue is the WHO Advisory Committee on Variola Virus Research, which oversees all research involving the last known stocks of smallpox virus. This advisory committee provides technical review and enforces an oversight framework that has maintained global trust despite the extreme risks posed by the variola virus. A similar body for mirror life–enabling research could create shared standards, offer politically neutral assessments, improve coordination among states, and reduce the likelihood that disagreement over science becomes a source of geopolitical tension.

Establishing such a committee requires extensive political agreement and sustained engagement from major scientific powers. If key states decline to participate, the committee's authority and deterrent value would be weakened. Some might also perceive such a body as vulnerable to political bias or becoming aligned with blocs, generating mistrust rather than reassurance. Finally, a failure by the committee to act credibly or transparently could undermine global restraint by signaling that oversight is ineffective.

Increase Domestic Economic Controls

Raising the economic and structural costs of pursuing mirror life within the United States is a flexible, early-stage deterrence lever that can reshape incentives without requiring criminalization or formal international agreements. These measures operate through existing governance systems and can be deployed incrementally. The measures function as a middle layer between informal norms and hard enforcement: They are more consequential than voluntary standards but less politically fraught than sanctions or criminal penalties. Options include

- **Funding restrictions:** Governments can restrict or deny public funding for research that advances the creation of mirror life. This lever does not prohibit such research outright but removes a major source of support for academic and institutional work. Precedents demonstrate how pauses in funding can slow high-risk research while governance frameworks are developed. After controversial experiments increased H5N1 avian influenza transmissibility in 2011, voluntary and government-imposed moratoria paused research while biosafety and oversight policies were strengthened (Fouchier et al., 2012). More recently, the U.S. government paused funding for a broader category of dangerous gain-of-function research pending a revised policy for high-consequence life sciences research (Administration for Strategic Preparedness and Response, 2014). Applied to mirror life, funding restrictions would slow technical progress by forcing researchers to rely on less-accessible private or foreign funding. They would also signal societal and governmental concern by placing mirror life work outside the public research mission. And they would create time for evidence-gathering and multilateral coordination before more-binding measures are considered. Because most universities and many research organizations rely on public funds, even a partial or temporary funding pause could meaningfully reshape research trajectories.
- **Liability and insurance:** Governments could impose a statutory strict-liability regime for research that would materially advance mirror life development, making actors responsible for the resulting harms, regardless of the level of care exercised. The purpose of such a regime would not be to ensure compensation after a catastrophe (because damages from a mirror life release could be effectively unbounded) but to influence behavior before harm occurs. Under strict liability, insurers would likely refuse coverage or price it prohibitively, as seen in the case of Superfund-related environmental liabilities. By making certain categories of enabling research legally and financially uninsurable, the regime would discourage institutions from pursuing work that carries unbounded downside risk, even in the absence of immediate harm.

Increasing economic barriers domestically carries several important risks and must be carefully tailored. Overly broad restrictions may sweep in benign or adjacent research, triggering backlash from scientific communities and others. Inconsistent international implementation also creates friction: If only a subset of major powers imposes economic constraints, actors could migrate to permissive jurisdictions, reducing visibility and oversight. Liability and insurance levers might disproportionately affect legitimate researchers while failing to deter malign actors who are resistant to financial pressure. Finally, without clear criteria, these tools risk being seen as arbitrary or ideologically motivated, undermining their credibility.

Despite these risks, economic barriers can play valuable roles when carefully designed: They are reversible, flexible, and capable of shaping incentives quickly while more-comprehensive governance measures are developed.

Restrict Access to Enabling Materials, Technologies, and Knowledge

Limiting access to the materials, tools, and knowledge needed to construct mirror life can slow development and reinforce global norms that mirror life is outside the bounds of responsible science. Some of these levers use existing governance mechanisms, making them comparatively fast to deploy. Such levers translate normative disapproval into structural impediments without requiring immediate global consensus:

- **Enact export controls on goods and technical information:** These restrict the cross-border transfer of sensitive goods, technologies, and technical data through licensing regimes that are administered by such agencies as the U.S. Department of Commerce under the Export Administration Regulations and the State Department under the International Traffic in Arms Regulations. Extending these controls to mirror life–relevant tools—specialized chemical reagents, mirror nucleotides, custom synthesis equipment, or computational design pipelines—would raise the cost of development and help align international expectations. This approach parallels how export controls have long shaped the proliferation landscape for nuclear technologies, cryptographic technologies, and advanced biotechnologies. However, export controls do not affect purely domestic transactions. If not paired with visible domestic restraint, export controls could fuel perceptions that the countries that impose controls are seeking unilateral advantage. This *us versus them* dynamic has been observed in biotechnology and AI export-control debates (Epstein, Crawford, and Nevo, 2025) and might hinder international consensus unless countries clearly demonstrate that they are constraining their own programs, as well as those in other countries.
- **Regulate key precursor materials and intermediates:** Restricting access to essential precursor materials provides a direct method for constraining capability. Such rules complement export controls by focusing on the **domestic control** of the raw inputs necessary to construct mirror biological systems. Potential precursors include mirror nucleotides, mirror amino acids, short mirror nucleic acids or mirror peptides, or reagents that enable nonstandard biochemical reactions. Potential intermediates include long stretches of mirror DNA, mirror proteins, mirror ribosomes, or other molecular machinery. Precedent for this approach is extensive. Governments have long controlled drug precursors, such as pseudoephedrine; many vendors of synthetic DNA screen their orders and their customers to minimize the probability that the DNA will be used in biological weapon development. And international agreements regulate chemicals used to produce chemical weapons (Giommoni, 2025; Williams and Kane, 2023). Controls can take the form of registration, licensing, tracking, bulk-purchase limits, inspections, or use-case declarations. Because precursor controls can be implemented unilaterally and expanded over time, they are flexible, scalable, and robust to uncertainty about when mirror life may become technically feasible. And unlike export controls, they also signal domestic restraint by meaningfully limiting one's own research pathways.

- **Control IP:** Controlling IP rights related to mirror life can reduce the financial incentives that motivate private firms, investors, and research institutions. States may impose secrecy orders on patent applications that raise national security concerns, as authorized in the United States under the U.S. Invention Secrecy Act of 1951 (Pub. L. 82-256, 1952). Such orders have historically restricted the dissemination of nuclear, cryptographic, and other sensitive technologies (Gross, 2019). It is possible that governments could go further by formally signaling that patents related to mirror organisms will not be granted or enforced. The United States may also have additional levers through Bayh–Dole march-in rights or licensing controls over government-funded research (Blevins and Hickey, 2024). A broad policy of nonrecognition would create legal uncertainty, undermining the commercial viability of mirror life technologies and deterring venture investment. Historical precedents suggest that such measures can have an effect: In 1934, physicist Leo Szilard patented nuclear chain reactions but assigned the patent to the British Admiralty to prevent misuse (Hargittai, 2023); during the Cold War, secrecy orders were used to restrict the dissemination of militarily relevant innovations.
- **Control publication:** Academic journals and professional societies serve as gatekeepers for scientific legitimacy. By refusing to publish research that creates or materially advances the development of mirror organisms, such gatekeepers can deprive practitioners of one of the main incentives for high-risk research: scientific recognition. Journals already exercise such discretion routinely, rejecting research that violates ethical norms, including work that is conducted without proper oversight or in ways inconsistent with accepted biosafety practices. The National Science Advisory Board for Biosecurity's intervention in the H5N1 publication debate illustrates both the influence and limitations of voluntary editorial controls (Fedson and Opal, 2013). If leading journals adopt consistent standards and coordinate across editorial boards, mirror life research could be starved of visibility, prestige, grant competitiveness, and career relevance. However, absent broad coordination, work may simply shift to less regulated venues. There are also risks of backlash from the scientific community or migration of research to jurisdictions with weak norms. Clear criteria, their transparent application, and periodic review can mitigate these risks.

These restrictions must be precisely scoped to avoid undermining legitimate research while effectively constraining activities that advance mirror organism development. Overly broad rules could impede beneficial science and provoke accusations of censorship, or they could push such research underground. Export controls and IP restrictions could be viewed as attempts to secure strategic advantages unless accompanied by unequivocal domestic restraint. Precursor controls must balance security and scientific progress to prevent unintended harm to biomedicine or biotechnology. Publication controls require broad editorial coordination to avoid fragmentation. And uneven adoption across countries could create loopholes that concentrate risk in permissive jurisdictions, weakening overall deterrence.

Criminalize and Prosecute Domestic Mirror Life Development

National legislation can be enacted to criminalize the creation of mirror life and any intermediates and precursors that are so enabling for or specialized to mirror life development that they should be banned as well. (Other substances or services that could facilitate mirror life production but that have other functions that need to be protected could be permitted under regulation, as described previously.) Criminalization would be established by Congress (or the respective national legislatures in other countries) and enforced by law enforcement agencies. Directly, this raises the costs and legal risks for domestic actors who might want to engage in mirror life R&D or supply others who are doing so, undermining both incentive and capability. Indirectly, it sends a chilling signal to investors and collaborators, thus discouraging them from financing or backing mirror life development.

One precedent for such criminalization is U.S. Code, Title 18, Section 175c, which bans the unauthorized possession or construction of the variola virus, which is responsible for smallpox (Epstein, Crawford, and Nevo, 2025). Similarly, Title VI of the Clean Air Act bans the production of CFCs, which pose high risk to the global environment. In that case, the ban occurred despite prior commercial interest (42 U.S.C. 85; Whitesides, 2020). Although criminalization requires prior legislative action and consensus, it provides a vital legal basis for other enforcement tools, such as surveillance or prosecution. However, to maintain international credibility, any such domestic law would have to make clear that any development of mirror life—including by officials of the state that enacts the legislation—would be illegal and that no such program could be legitimately authorized. Other risks include potential backlash over scientific freedom or the displacement of research to unregulated jurisdictions. Effectiveness hinges on robust enforcement and jurisdictional reach.

Levers to Disrupt or Punish Advanced Development

The levers in this category represent the most forceful and highest-regret tools available for deterring actors from pursuing mirror life. *Disruption* involves physically or digitally preventing the construction of mirror organisms when other means of dissuasion or prevention are unavailable or have failed. *Punishment* entails imposing legal, diplomatic, or material consequences on actors who cross defined boundaries. Their deterrent effects arise from both the capability and the credible willingness to act (Mazarr, 2018).

Disruption and punishment are familiar elements of other deterrence and nonproliferation strategies. However, as we discuss in Chapter 5, mirror life presents a fundamentally different challenge, and these tools are likely to undermine the cooperation on which prevention depends.

Nevertheless, disruption and punishment might still enter policy discussions. These levers are therefore discussed here only to clarify their limitations and to underscore why the existing strategy focuses instead on building the legitimacy and collective capacity required for universal restraint.

Impose Sanctions and Economic Isolation

Treasury departments, foreign ministries, and international organizations, such as the UN Security Council, can impose sanctions targeting individuals, institutions, firms, or governments suspected of advancing mirror life. Sanctions raise direct financial and logistical costs for the targets while indirectly deterring affiliated private-sector actors and diplomatic partners. Historical examples include sanctions on Iran's nuclear program and Russia's use of chemical weapons (Masterson, 2021; Sen, 2018). The success of these actions depends on accurate detection and coordinated enforcement, which is one reason that recent sanctions on Russia have failed (Spencer, 2025). Although sanctions can sometimes create leverage for compliance or negotiations, they risk entrenching resistance or provoking countermeasures, which are dangerous responses in the mirror life domain.

Apply Visa and Academic Restrictions

Academic visa restrictions can be used to deny travel privileges to researchers who are affiliated with institutions that conduct impermissible research. Immigration authorities and foreign ministries control these measures, which directly impede access to global academic networks and indirectly isolate institutions. Although such restrictions create diplomatic and legal pressure, they risk political backlash and unintended targeting.

Operational Disruption

Operational disruption aims to delay or degrade development when prior deterrence measures have failed. Disruption can directly weaken capabilities by sabotaging infrastructure or materials and indirectly alter incentives by signaling that attempts to create mirror life will face active resistance. Disruption can refer to several types of actions:

- **Supply-chain disruption.** Governments can interfere with an actor's access to materials or equipment required for mirror life development by interdicting shipments, introducing defects, or embedding surveillance or fail-safe mechanisms into critical components.
- **Cyber disruption.** Cyber operations can sabotage mirror life programs by corrupting data and damaging digital control systems or by altering the computational models used in protein engineering or ribosome design. These actions may be subtle—such as data manipulation that causes erroneous lab results—or overt, such as disabling infrastructure altogether.
- **Physical disruption.** States have considered and exercised covert or overt kinetic actions to destroy facilities, equipment, or materials associated with catastrophic weapons. In the mirror life context, physical disruption poses unique dangers: If mirror organisms are already present, a strike could inadvertently disperse them.

Although such actions might appear to offer a way to halt technological progress, they carry technical, diplomatic, and strategic risks of inadvertently leading to the creation or release of mirror life. Physical strikes risk international condemnation, collateral damage, or the dispersal of hazardous materials if the target contains mirror life, and the threat of disruption could strain relationships and trigger the very race toward mirror life that such activities seek to deter.

Levers to Deploy Countermeasures and Build Resilience

Countermeasures and resilience are central features of traditional biodefense planning and threat planning in general. However, mirror life presents a fundamentally different challenge. Unlike conventional pathogens—for which vaccines, therapeutics, or environmental interventions can sometimes mitigate harm—no plausible countermeasure appears likely to contain or neutralize mirror life at the scale required once it is released (Adamala, Agashe, Binder, et al., 2024). Any pharmacological protection would have to be administered continually and globally: Environmental countermeasures would face an adversary capable of colonizing the entire biosphere, and no existing public-health or biodefense model has ever succeeded in deploying a countermeasure at a planetary scale. For these reasons, prevention remains the only reliable defense.

Nevertheless, countermeasure research and resilience planning may still enter policy discussions. In traditional threat domains, defensive preparations can strengthen deterrence—for example, the vaccination of select military forces historically deterred potential adversaries from employing smallpox as a weapon. Yet, for mirror life, countermeasure development carries distinct pitfalls. Developing defensive tools might bring research closer to the creation of mirror organisms; defensive programs could be misconstrued as offensive programs or as preparations for future offensive programs, triggering competitive spirals; and most countermeasures are unlikely to adequately protect either human populations or global ecosystems. Still, the carefully governed exploration of defensive concepts, especially through transparent international forums, might help reduce uncertainty and prevent destabilizing unilateral programs.

Medical and Environmental Countermeasures

Medical and environmental countermeasures aim to prevent or reverse harm from a mirror life release. Although these approaches are highly constrained by biology and feasibility, they may be explored conceptually to understand their value and limits. Because countermeasure development could unintentionally advance research that would enable mirror life, any such work would require extraordinary oversight and strong guardrails, potentially through international transparency mechanisms, such as WHO-style committees or Coalition for Epidemic Preparedness Innovations–like cooperative R&D models (Hay and McCauley, 2018; Halabi et al., 2023). Potential options for countermeasures include the following:

- **Antibiotics and therapeutic agents:** Some nonchiral or racemic (mixed chirality) antibiotics might be active against mirror bacteria, and additional pharmacological agents could theoretically be engineered using known biochemical principles (Adamala, Agashe, Binder, et al., 2024). However, efficacy cannot be rigorously tested without access to mirror bacteria or mirror bacteria–immune-system interactions, limiting scientific confidence in the agents' performance. Untested mirror-oriented therapeutics also could have unpredictable effects on human immune systems. Even if partially effective, any antibiotic campaign would protect humans but not species outside direct human care or control, leaving natural ecosystems and food webs vulnerable to collapse (Adamala, Agashe, Binder, et al., 2024).
- **Vaccines:** Vaccines based solely on mirror biomolecules would likely produce little or no protection because mirror molecules would not be processed effectively by natural-chirality immune systems (Adamala, Agashe, Binder, et al., 2024). *Conjugate vaccines*, which attach mirror antigens to natural-chirality carriers, might elicit antibody responses but would require frequent boosting and enormous mirror-molecule manufacturing capacity. As with antibiotics, even a highly effective human vaccine could not protect nonhuman species or global ecosystems.
- **Mirror bacteriophage (phage):** One proposed environmental countermeasure is developing a mirror phage capable of infecting and killing mirror bacteria (Adamala, Agashe, Binder, et al., 2024). Creating a mirror phage might be easier than constructing a mirror bacterium, but evolutionary theory predicts that mirror bacteria would rapidly become resistant to such phages (Bohannan and Lenski, 2000). No ecological precedent suggests that a phage could reverse a global-scale invasion once mirror bacteria had established themselves. Additional research would be needed to understand whether any environmental intervention could meaningfully slow ecological collapse.

Countermeasure research carries several significant risks. First, it might inadvertently accelerate progress toward mirror life. Developing realistic defensive tools may require the use of mirror-like molecular systems or experimental surrogates or the creation of computational simulations that lower the barriers to creating mirror organisms. Second, countermeasure programs could be misinterpreted as either hedging or covert offensive research. In a low-trust geopolitical environment, even purely defensive preparations can appear destabilizing—much as ballistic missile defense was seen during the Cold War (Kimball and Reif, 2020)—potentially prompting others to advance their own efforts toward creating mirror life. Finally, countermeasures offer limited practical value. Even in the most optimistic scenarios, they are unlikely to prevent global ecological collapse or widespread mortality if mirror organisms were released. Overinvesting in countermeasures risks creating false confidence, distracting attention from prevention and toward strategies that cannot substitute for early restraint.

Societal Resilience

Resilience focuses on prolonging life and sustaining essential services, as well as slowing collapse in the event of a mirror life release. Unlike countermeasures, resilience does not aim to neutralize mirror organisms; instead, resilience seeks to protect populations temporarily or reduce cascading societal failures. However, the extraordinary scope of mirror life threats limits the feasibility of long-term resilience. A mirror life release could undermine agriculture and food webs, rendering most conventional emergency planning (e.g., stockpiles, personal protective equipment [PPE], sheltering, supply-chain continuity) insufficient.

The following approaches are theoretical:

- **PPE and containment practices:** Enhanced PPE could protect individuals temporarily, delaying infection or colonization. However, even highly effective PPE cannot support long-term survival in the face of ecosystem collapse. Protective measures might slow mortality but could not prevent eventual resource scarcity.
- **Medical prophylaxis stockpiles:** Antibiotics or prophylactic drugs could be stockpiled for temporary protection of critical personnel or vulnerable populations. Although this approach is limited in effectiveness, such stockpiles might extend survival or support emergency response operations for short periods.
- **Agricultural and ecological mitigation:** Interventions, such as anti–mirror organism pesticides, temporary greenhouse systems, or controlled-environment agriculture, might protect select crops or animal populations. However, these strategies require sustained effort, which would likely fail under widespread ecological disruption.

Resilience measures also carry several risks. Most fundamentally, they might create a false sense of security, implying that survival after a mirror life release is possible when, in reality, no emergency response (e.g., protective gear, food-system intervention) could sustain human or ecological systems if the release of mirror life led to global biome collapse. Even well-intentioned preparedness efforts can therefore weaken the political case for prevention by suggesting that catastrophic outcomes are manageable. Resilience programs also risk being misinterpreted as forms of offensive hedging. In a low-trust environment, resilience investments could be perceived as preparations to survive one's own release or to endure a rival's, prompting others to invest rather than desist in developing mirror life. Finally, the practical utility of resilience measures is extraordinarily limited. No conceivable strategy could maintain long-term human survival, much less ecological stability, after the widespread establishment of mirror organisms. For these reasons, any resilience planning must be modest and transparently framed as contingency preparedness to support prevention, *not* as a viable fallback should prevention fail.

CHAPTER 5

Principles for a Mirror Life Strategy

Mirror life presents a strategic problem unlike any other global threat in human history. Nuclear weapons, pandemics, invasive species, and climate change have all produced visible warning signs that galvanized public awareness, built shared understanding of the risks, and allowed governance to evolve in response.

Mirror life may offer no such opportunity. The moment a viable mirror organism is created might also be the moment that humanity loses the ability to contain or counter it. This is not because the first mirror cell would be inherently dangerous. It is because creating a viable mirror organism is the steepest technical hurdle, and once cleared, the path to engineering more-dangerous forms becomes far easier.

Yet this same structure of risk contains an unexpected silver lining. Mirror life has not yet been created, and its development would require extraordinary effort and investment. Only a small number of scientifically advanced states possess—or will soon possess—the capacity to create mirror life independently. Establishing shared norms and agreements among these powers is therefore not only necessary but, crucially, *sufficient* to prevent mirror life's creation for the foreseeable future.

The challenge, then, is to create a strategy that enables capable actors—many of whom mistrust one another and compete for scientific or strategic advantages in other arenas—to recognize their common interests and establish shared restraint on this issue.

In this chapter, we first elaborate why mirror life differs fundamentally from historical analogues. We then translate the structural features of the problem into the following four principles that should guide a strategy aimed at preventing the creation of mirror life:

1. **Prevention over mitigation:** Action occurs before the first mirror organism exists because creating the first mirror cell might be the point at which containment and defense become impossible.
2. **Universal restraint over partial restraint:** All actors are deterred from creating mirror life because creation of mirror life by any actor can pose a global catastrophic risk.
3. **Transparency over secrecy:** Transparency promotes the trust and collaboration needed to prevent the creation of mirror life, whereas secrecy is infeasible and strategically destabilizing.
4. **Cooperation over coercion:** Preventing the creation of mirror life requires preserving the world's capacity for cooperation. Because only a few major scientific powers can create mirror life independently, their restraint is essential to global safety. This means prioritizing cooperation among these powers, including traditional competitors, and avoiding the use of tools that erode trust.

Together, these principles form the conceptual foundation for a global strategy to prevent the creation of mirror life. These principles differ sharply from the principles that have governed other global risks because mirror life is not a problem to be managed but a catastrophe to be avoided outright.

Mirror Life in Context: Why Historical Analogues Fail

Decisionmakers have grappled with global threats throughout modern history, from nuclear weapons to pandemics to climate change. The historical efforts to counter these threats offer mental analogues and lessons that might initially seem relevant to the challenge of mirror life. Defense experts might recall WMDs, technologists might reference dual-use research, environmental regulators might point to invasive species or climate stressors, and natural scientists might cite pandemics or asteroid strikes.

However, mirror life has a novel combination of characteristics that renders these analogues incomplete. Mirror amino acids and mirror nucleic acids—mirror life's precursor materials—are widely available, inexpensive, and deeply embedded in global chemical and biotechnology supply chains, making risk mitigation strategies of regulating precursor materials (such as regulating uranium to reduce the risk of nuclear weapon development) largely impractical. The potential effects of mirror life are planetary and irreversible, meaning that strategies that rely on containment, partial risk reduction, or resilience and survival would likely be inadequate. And mirror life's creation would rapidly proliferate the knowledge necessary to produce more-robust and more-dangerous forms, rendering ineffective those policies solely focused on controlling access to or restricting information. Taken together, these factors demand a policy framework that diverges sharply from past experience. In the remainder of this chapter, we develop that framework.

Principle 1: Prevention over Mitigation

In Chapter 1, we argued that mirror life requires **preemptive governance**: action before the risk is realized. Here, we justify this principle of **prevention** over **mitigation** more fully.

Policymakers typically have the benefit of time and exposure in confronting catastrophic risks. The ozone hole, nuclear testing, the first major oil spills, and climate-fueled disasters, to name a few, provided visible crises that galvanized public concern and clarified the need and knowledge necessary for regulation. Even rapidly emerging technologies, such as generative AI, offer real-world experience and early warning signs with which to make policy.

Mirror life presents a fundamentally different challenge. The creation of the first viable mirror cell could represent a point of no return because once this technical barrier is crossed, as we describe in the following paragraphs, containing it or mitigating its harms might not be possible and the knowledge and methods for creating more-robust or more-dangerous strains would likely proliferate rapidly.

Thus, waiting for evidence of the harm of mirror life before acting would be tantamount to accepting irreversible loss because of the ineffectiveness and dangerousness of containment and defense options. The only viable window for governance is before mirror life exists. Acting now allows policymakers to establish norms and constraints that prevent the first creation of mirror organisms rather than attempt to manage their consequences after control has been lost.

Fortunately, mirror life remains hypothetical: It exists only in the minds of scientists, and it will not become a reality unless humans choose to make it so and work hard to do so. **This means that the scientific and policy communities have the rare opportunity to study and develop governance mechanisms that could address the threat from mirror life before it is realized**, and those mechanisms can be implemented before losses occur. This means that the scientific and policy communities have a rare opportunity to study, develop, and implement governance mechanisms that would prevent the creation of mirror life before it is possible to do so. Early governance could also stop significant investments in technologies that would enable the creation of mirror life, thereby avoiding the costly disruptions, stranded assets, and resistance that often accompany regulatory changes that are introduced after an industry is already established.

Failures of Containment

A mirror life strategy should prioritize prevention over mitigation because of likely failures in containing either the mirror organism itself or the tools and methods needed to create it. Moreover, not all scientists in laboratories doing such research can be guaranteed to have and to persist in having benign intent. Creating any mirror organism—even one designed to be harmless or strictly dependent on laboratory conditions—would provide the knowledge and tools that other actors could use to engineer more-dangerous or more–environmentally robust mirror strains. Even a "safe" mirror organism could therefore enable the spread of capabilities needed to develop forms of mirror life that pose far greater risks, including the environmentally persistent mirror bacteria described by Adamala, Agashe, Belkaid, et al. (2024). Containment also cannot prevent the diffusion of tacit expertise, design methods, or genomic constructs that could be repurposed for harmful ends.

Physical containment systems themselves are imperfect and prone to failure or subversion. Even if scientists pursue mirror life with benign intent, biocontainment measures cannot guarantee confinement. Historical experience shows that accidental exposures to dangerous pathogens have occurred even in the world's most advanced high-containment laboratories (Günther et al., 2011; Kortepeter et al., 2008; Miller, 2004). If such failures occur with natural pathogens in facilities built for them, there is little reason to expect that there would be perfect containment of a novel form of biology.

Biological containment is a strategy that seeks to engineer features into an organism that make it impossible for the organism to survive and propagate outside the laboratory. Even if robust biological containment of mirror organisms were possible—containment that would reliably resist continued evolution of the organism once it is released—any such features could always be engineered out. In short, biocontainment measures within laboratories would not be able to reliably prevent uncontained mirror organisms from being created.

Environmental containment is equally unlikely. In the natural world, bacterial populations are constrained by predation, viral infection (bacteriophages), and competition; mirror bacteria might be highly resistant to these controls. Even if mirror bacteria's intrinsic growth rate were slower than that of natural bacteria, the absence of mortality pressures could yield a positive net growth rate, enabling the bacteria's spread through the environment and living hosts. Once established, mirror organisms would evolve like any other life form. Mirror bacteria could consume achiral nutrients from the outset, and subsequent mutations could expand their metabolic range to include natural-chirality nutrients, increasing their robustness and survivability outside laboratory conditions. If engineered to metabolize natural sugars, mirror bacteria could become even more persistent. Environmental spread could cause fatal infections in humans and other organisms and destabilize ecosystems and agriculture at scale.

Failures of Mitigation

Mirror bacteria would be composed of biomolecules with opposite chirality from natural life. Because immune systems recognize pathogens by binding to specific molecular structures, mirror molecules would likely fail to trigger immune activation, which would impair many immune responses. Mirror bacteria could proliferate widely within human hosts, producing sepsis-like illness or organ disruption.

As described in Chapter 4, governments or research institutions might attempt to design medical treatments, environmental controls, or biological defenses in advance, but these mitigation efforts would face formidable challenges. Protective measures would need to apply not only to humans but also to agricultural systems and ecosystems. Even if highly effective medical countermeasures for mirror bacterial infections could be developed, they would likely be unevenly distributed among human populations, leaving many unprotected. Even if a government could protect its own population from the effects of mirror life release, other countries might suffer significant effects, which could potentially escalate conflict. And deploying medical countermeasures at the scale required to ensure survival of ecosystems could be entirely impossible.

Moreover, mirror bacteria would undergo evolutionary adaptation like any other living species, eroding the effectiveness of any single intervention over time, just as antibiotic resistance emerges in natural bacteria. The coronavirus disease 2019 (COVID-19) pandemic illustrates this challenge: Despite the unprecedented mobilization of resources when the pandemic emerged, vaccines and treatments were not delivered universally, and evolution produced new variants over time. A mirror life release would pose an orders-of-magnitude-greater challenge.

Countermeasure development could also be strategically destabilizing. Many states have pursued defensive capabilities to protect themselves against naturally occurring and intentional biological threats. Some actors might believe it would be possible to create countermeasures against mirror life—developed without creating mirror life itself—that could mitigate some of the anticipated effects of mirror life. However, if one actor mistakenly believes that it can survive or neutralize a mirror life release, it might be more willing to pursue mirror life for itself or tolerate its development by others. This could be destabilizing in the following two ways:

- It might embolden some actors to develop mirror life for coercive use.
- Others might misinterpret signs of countermeasure development as a cover for mirror life development or a belief that mirror life release might be survivable, undermining trust and escalating competition.

Principle 2: Universal Restraint over Partial Restraint

A mirror life strategy should prioritize **universal restraint** over **partial restraint**—constraining all, not most, actors from creating mirror life. Although a strategy of partial restraint might significantly reduce risks in other domains—e.g., nuclear nonproliferation, biological weapon controls, climate agreements, AI safety—partial restraint would be largely ineffective for mirror life because even a single state or nonstate actor could trigger irreversible global ecological consequences. Partial restraint therefore reduces risk only marginally.

A universal-restraint approach, by contrast, aims to deter or dissuade all actors from creating mirror life, reflecting the biological reality that a mirror organism created anywhere could become a global problem. Although universal restraint is more demanding, it aligns more closely with mirror life's risk structure and provides a more stable basis for long-term prevention. A strategy to prevent mirror life would thus need to incorporate how governance arrangements, transparency mechanisms, diplomatic engagement, and scientific norms can work together to support universal restraint.

Principle 3: Transparency over Secrecy

A mirror life strategy should prioritize **transparency** over **secrecy**. Both transparency and secrecy approaches have precedents in past efforts to govern dangerous technologies, and both have advantages and disadvantages. In the mirror life context, however, a central factor in determining which strategy will work is how that strategy shapes global perceptions of risk.

Because mirror life risks cannot be demonstrated empirically without creating the threat itself, prevention depends on how actors interpret scientific assessments and whether the actors trust one another's intentions. When actors share an understanding that mirror life poses severe dangers, collective restraint becomes possible. Rational actors who accept the scientific warnings are unlikely to pursue mirror life because any attempt to use it would almost certainly harm the user to a degree that would negate any conceivable benefit. When risk perceptions diverge, information gaps may still be bridgeable:

- An actor unaware of mirror life's dangers might change course when presented with credible evidence.
- An actor convinced that mirror life is safe would have strong incentives to share that evidence openly to avoid misperception by others.

If agreement on the risks is achieved, incentives to pursue mirror life might disappear. But persistent uncertainty or disagreement, especially if combined with beliefs that mirror life might yield strategic or scientific gains, could trigger competitive dynamics. Some actors might believe that mirror life could enable breakthroughs in biotechnology and might assume that other actors will race to exploit those benefits, even though many scientists believe that most benefits of mirror biology can be achieved without creating viable mirror cells (Adamala, Agashe, Binder, et al., 2024). These perceptions could produce a fear of falling behind, prestige-seeking, or the belief that development is inevitable. In this way, risk perception itself can drive pursuit. The choice between secrecy and transparency must therefore be evaluated according to how each strategy influences these perceptions.

A suppression-and-secrecy strategy might rely on covert restrictions on research and undisclosed monitoring of potential development efforts. This model has historical precedent in the conduct of classified nuclear and intelligence programs during the Cold War (Wellerstein, 2021). In theory, suppression could delay proliferation long enough for diplomatic coalitions to form.

However, in the mirror life context, secrecy could distort how actors perceive risk, potentially making the situation less stable and triggering arms-race dynamics. When intentions cannot be observed, actors might infer that others are making secret progress, especially in a domain in which risks cannot be empirically demonstrated and where potential benefits, however speculative, may, to some, appear significant. Secrecy also prevents actors from building a shared scientific understanding of mirror life's dangers, which must be conveyed through reasoning and open analysis rather than empirical demonstration. Secrecy can further undermine verification by forcing actors to choose between intrusive monitoring and ineffective oversight (Vaynman, 2021).

Secrecy can also not be assured indefinitely, as illustrated by the long history of espionage and information leakage. Furthermore, secrecy cannot meaningfully restrict access to enabling materials, such as mirror amino acids and nucleotides, which can already be synthesized with globally available technologies (Diercks, Dik, and Schultz, 2021; Harrison et al., 2023; Katoh, Tajima, and Suga, 2017; Pollegioni, Rosini, and Molla, 2020; Young, Kundu, and Sczepanski, 2019). Unlike fissile materials (Glasgow, Teplinsky, and Markus, 2012), these cannot be tightly controlled. As a result, secrecy increases suspicion without effectively constraining capability.

Transparency shapes risk perception in more-stabilizing ways. By openly sharing scientific assessments of mirror life's dangers, states can converge on the understanding that pursuing mirror life would be self-destructive for any rational actor because its ecological spread would threaten both the initiator and everyone else (Adamala, Agashe, Binder, et al., 2024). Transparency can

- align risk perceptions, reducing the chance that actors underestimate risks or overestimate benefits
- reduce misinterpretation, making it harder for actors to view defensive research or risk-reduction measures as offensive preparations
- support durable norms that delegitimize the pursuit of mirror life
- clarify the true threat, distinguishing responsible scientific communities from isolated and sanctioned states or omnicidal actors.

Historical precedents reinforce this logic: The Asilomar Conference on recombinant DNA (Berg, 2004), the Montreal Protocol (Manzer, 1990), and transparency mechanisms in arms-control treaties (Zarimpas, 2003) all illustrate how openness reduces incentives for arms races and strengthens cooperation.

Viewed through the lens of risk perception, transparency is not only preferable but might be strategically necessary. Suppression cannot meaningfully restrict access to enabling materials or knowledge but does intensify mistrust. By contrast, transparency fosters aligned perceptions of risk and strengthens the conditions for collective restraint.

Principle 4: Cooperation over Coercion

Finally, a mirror life strategy should prioritize **cooperation** over **coercion**. Individually, every country has strong incentives to prevent the domestic development of mirror life because of the lack of benefit and catastrophic potential. However, domestic action alone cannot fully manage the global risks of mirror life, and mistrust or gaps in governance between countries could still allow mirror life to emerge elsewhere. Therefore, the first obligation of any prevention strategy is to preserve the legitimacy and trust that make effective collective action possible. This has at least two implications: pursuing inclusive leadership and foregoing coercive tools.

Pursuing Inclusive Leadership

A mirror life strategy should aim for inclusive leadership to manage mirror life governance. Coordinating first with close allies (e.g., participants in the Australia Group, the PSI, or NATO) might appear to be a natural and convenient starting point. These groups have established procedures, trusted relationships, and proven capacities for cooperation on other biological and technological risks. But for mirror life, beginning with "friends first" is not only insufficient, it can be destabilizing.

A tempting analogy for coalition building is the Cold War model: Cultivate shared deterrence or stability with allies and manage competitors through strength. But mirror life does not behave like other technologies for which adversaries can engage in tit-for-tat deterrence.

This has direct implications for coalition design. If governance begins within exclusive blocs, it can incorrectly signal that mirror life is a domain for strategic advantage or military leverage. That signal alone can reinforce the very misperceptions that could provoke a race. Actors excluded from allied processes might fear being left behind or suspect that mirror life is being explored for competitive gain, even though no rational state could use it safely.

Ally-centric strategies also face several structural weaknesses that mirror life uniquely amplifies. First, these strategies exclude major scientific powers whose participation is essential. China, Russia, and other technologically capable states are not members of the Australia Group, the PSI, or NATO. Because mirror life precursor materials are globally available, exclusion cannot constrain capability; it merely shifts activity outside the governance framework.

Second, exclusion breeds suspicion and fuels competitive dynamics. Without global participation, others might interpret allied governance efforts as attempts to consolidate strategic advantage, especially if countermeasure R&D or monitoring tools are involved. In a domain in which risk perception is itself a driver of danger, exclusion magnifies misinterpretation.

Third, ally-first approaches require contentious political reinterpretations. Extending existing chemical or biological control lists to mirror life could require classifying mirror organisms as WMDs. Such moves risk deepening geopolitical divisions, entrenching competitive narratives rather than preventing them.

Bringing all major scientific powers into governance from the outset offers the following key advantages:

- **Reduced misperception:** Common forums decrease the likelihood of worst-case assumptions.
- **Safer countermeasure development:** Inclusive platforms allow defensive R&D to be conducted transparently, thus reducing suspicion.
- **Realistic capabilities:** Because mirror life precursors are widely accessible, only inclusive governance can meaningfully constrain development.

Forgoing Coercive Tools

Coercive tools face practical and ethical limitations. Attribution of responsibility for mirror-life research would be uncertain, targets could be misidentified, and the effects of disruption could escape intended boundaries, harming legitimate scientific activity. These risks, combined with the potential to undermine emerging consensus, make such measures out of scope.

By contrast, a strategy grounded in transparency, dialogue, and shared legitimacy strengthens the conditions for restraint. Building cooperative capacity among the major scientific powers is both the most effective and the most stable path to prevention.

Conclusion

Preventing mirror life requires more than isolated policies or technical controls: It demands a coherent governance approach shaped by preemptive action, universal reach, transparency, and cooperation. These **strategic choices** establish the foundation for any credible effort to prevent the first creation of mirror organisms, and they frame the central policy questions that follow in Chapter 6: *What* actions are available to deter the creation of mirror organisms and *how* can these actions be assembled into a stable and effective U.S. strategy?

CHAPTER 6

A U.S. Strategy for Mirror Life

In this chapter, we describe a U.S. strategy to help shape a global environment in which restraint from creating mirror life is the agreed norm among all actors with the capability to do so. The strategy applies the principles developed in Chapter 5—prevention over mitigation, universal restraint over partial restraint, transparency over secrecy, and cooperation over coercion—and organizes the levers from Chapter 4 into two complementary lines of effort (LOEs). Each LOE (summarized in Table 6.1) draws on two main categories of levers from Chapter 4: reducing incentives and detecting and deterring early-stage development. The LOEs are as follows:

- LOE 1 engages scientific superpowers—especially China—to build the foundation for collective restraint. This LOE draws on primarily voluntary, nonbinding levers that reduce incentives to pursue mirror life: specifically, improved scientific assessment and early norm-setting. Consistent with strategic principles, this approach emphasizes transparent collaboration among the actors whose biotechnology ecosystems and geopolitical weight are most capable of shaping global behavior. U.S.-China alignment is especially important because these countries' choices have the greatest potential to trigger or avert competitive dynamics.
- LOE 2 aims to establish collective restraint at three levels: domestically, within the United States; among other scientifically advanced states, including China; and across a broader set of legitimate actors, such as commercial firms, academic research institutions, and other nation-states. This LOE combines stronger incentive-reducing levers with levers that detect and deter early-stage development, including monitoring and cost-raising measures that jointly shift incentives and capabilities.

Although these LOEs build on one another, many actions can proceed in parallel, particularly those that require long lead times, such as domestic legislation, the development of detection technologies, or the building of treaty infrastructure. Because the risks associated with mirror life are both severe and uncertain, the strategy is designed to be adaptive to changing conditions. For example, at present, the strategy recommends forgoing countermeasure development because of its likely ineffectiveness and destabilizing potential while also outlining circumstances under which this judgment may need to be revisited.

Across all LOEs, the objective is the same: Create a world in which every capable actor chooses not to pursue the development of mirror life.

Line of Effort 1: Engage Scientific Superpowers to Build the Foundation for Collective Restraint

The first LOE seeks to create the underlying conditions necessary for global restraint from developing mirror life. Early action is important because risk perception among major scientific powers—especially the United States and China—can strongly influence whether mirror life is viewed as a shared global hazard or a com-

TABLE 6.1
Summary of a Possible U.S. Strategy for Preventing Mirror Life

	LOE 1: Engage Scientific Superpowers to Build the Foundation for Collective Restraint	LOE 2: Establish Collective Restraint
Objective	Shape early expectations by improving scientific understanding, aligning interpretations of risk among major powers, and establishing shared norms that make restraint the natural baseline.	Translate emerging consensus into durable domestic, bilateral, and multilateral constraints that raise the costs and lower the incentives of pursuing mirror life. Build monitoring architectures for credible enforcement.
Levers implemented	• Conduct additional risk assessments to address remaining uncertainties. • Research safer alternatives to mirror life. • Develop voluntary and ethical standards. • Issue public and multilateral statements and resolutions.	• Detect and deter early-stage development. • Expand monitoring. • Develop verification methods. • Establish a global scientific oversight committee. • Increase direct economic barriers. • Restrict access to enabling materials, technologies, and knowledge. • Criminalize and prosecute domestic mirror life development.
Actors affected and unaffected	• Influences the incentives but not the capabilities of scientifically advanced states, academic institutions, commercial firms, and most nation-states • Has little direct effect and limited indirect effect on isolated and sanctioned states and on extremist groups or lone individuals	• Shapes both incentives and capabilities of all actors currently capable of creating mirror life, including scientifically advanced states, commercial entities, and academic institutions • Directly, partially constrains isolated and sanctioned states through surveillance, controls, and diplomacy. Has limited direct impact on fringe extremist actors. Indirectly constrains these actors to the extent that capable actors desist

petitive technological frontier. Transparent, collaborative engagement among these powers reduces the likelihood that scientific advances or policy initiatives will be misinterpreted as hedging or covert development, which could otherwise accelerate competitive dynamics.

LOE 1 therefore aims to establish a shared scientific foundation and convergent norms that make restraint the natural and widely endorsed position. These actions are low-regret, reversible, and broadly defensible, and they form the basis on which the more-demanding levers in LOE 2 depend.

Levers and How They Work

LOE 1 focuses on voluntary and nonbinding levers that reduce the incentives to develop mirror life. This includes levers to

- **Conduct additional research and risk assessments to address remaining uncertainties:** This includes research on health and ecological hazards, detection methods for incremental progress, and potential choke points.
- **Research alternatives to mirror life and the technologies that would enable its creation:** Alternatives can lower incentives for commercial or academic actors who might otherwise pursue mirror life for biomedical, biomanufacturing, or scientific applications.

Collaborative implementation could involve parallel U.S.-China risk assessments, reciprocal scientific briefings, shared workshops among national academies, and coordinated research agendas on safer alternatives. These activities would allow each side to understand how the other interprets the science. These activities would also demonstrate mutual commitment to precaution rather than competition. Broadly, such scientist-to-scientist engagement could provide an alternative channel for building shared risk understanding and norms of professional self-restraint, particularly in relationships for which formal diplomatic coordination may be constrained.

Pursuing these knowledge-advancing levers first would strengthen credibility and an evidentiary basis for subsequent messaging and norm-building actions:

- **Develop voluntary and ethical standards.** These can define responsible scientific conduct and categorize mirror life–enabling research as ethically unacceptable.
- **Issue public and multilateral statements and resolutions.** Such messaging should emphasize that no scientifically supported benefits outweigh the risks of creating a self-replicating mirror organism and that even the strictest biosafety systems cannot guarantee perfect containment.

Collaborative implementation could include harmonized statements by U.S. and Chinese scientific academies, the coordinated publication of ethical frameworks, joint participation in WHO or UN workshops, and synchronized national policy signals that mirror life development will not be publicly funded. Some actions, such as the pursuit of international research moratoria, could involve substantial coordination with nongovernmental bodies, such as scholarly societies and industry consortia.

Actors Affected

LOE 1 primarily shapes incentives rather than capabilities. Its influence varies across actor categories.

- **Major scientific powers:** These actors are central to LOE 1 because their cooperation or competition sets global expectations. Collaborative scientific assessments and coordinated normative signals can help prevent competitive dynamics among them and make restraint more credible to others.
- **Other nation-states:** Other states tend to follow cues from major scientific powers. Visible consensus among the United States, China, and other scientifically advanced states strengthens the legitimacy of restraint and helps diffuse global norms.
- **Academic institutions:** Universities and scientific societies are sensitive to scientific consensus, professional ethics, funding signals, and reputational considerations. They are strongly influenced by LOE 1's levers.
- **Commercial and private-sector actors:** Commercial firms typically respond to regulatory expectations, investor sentiment, and liability frameworks. Early public norms or coordinated statements from scientific powers can signal future policy directions, thus encouraging firms to focus on safer research paths.
- **Isolated or sanctioned states, extremist groups, and lone actors:** These actors are the least influenced by norms but might still be affected indirectly. Clear scientific assessments and shared public norms can undermine narratives that mirror life might offer some value. If mirror life becomes universally stigmatized and viewed as extremely risky, even malign actors face increased reputational and operational barriers.

Risks and Potential Pitfalls

Although LOE 1 relies on low-regret tools, the following risks warrant attention:

1. **Misinterpretation of scientific work:** Even benign research (e.g., exploring mirror precursors or alternatives) could be misread as covert hedging if it is not conducted transparently. Collaborative communication among major powers helps mitigate but cannot eliminate this risk.
2. **Reputational or political costs of unsuccessful norm-setting:** If efforts to establish norms or moratoria fail, the United States could face criticism or lose credibility. However, such costs are outweighed by the benefits of attempting early alignment.
3. **Potential to alert omnicidal actors:** Public discussion of catastrophic risks could draw attention from actors who are seeking mass-harm tools. However, withholding information would disadvantage legitimate actors and is unlikely to meaningfully affect malign ones.
4. **Overly restrictive or uneven norms pushing research offshore:** If major powers' norms diverge sharply from each other, research might shift to jurisdictions with weaker oversight. This risk reinforces the importance of inclusive, collaborative norm-setting.
5. **Premature pursuit of moratoria polarizing scientific communities:** If scientific uncertainty remains high, strong calls for moratoria could provoke resistance. Early joint assessments and transparent scientific dialogue reduce this risk.

Line of Effort 2: Establish Collective Restraint

The second LOE seeks to translate the scientific and normative foundations that were established in LOE 1 into sustained collective restraint with levers that further reduce incentives and constrain capabilities to develop mirror life. It also begins building an early detection and monitoring architecture.

Where LOE 1 shapes expectations, LOE 2 shapes the institutional arrangements, legal frameworks, and material conditions that make mirror life development costly or infeasible for a broad set of actors. Because mirror life is a global risk made possible by widely accessible enabling technologies, collective restraint is more durable when the most–scientifically capable states—including the United States, China, the European Union, the United Kingdom, and Japan—signal restraint together. Their alignment helps prevent competitive dynamics and lays the foundation for a broader global equilibrium.

Levers and How They Work

The first set of LOE 2 levers, in the list that follows, strengthens collective restraint by establishing shared expectations and baselines for behavior, making it harder for any actor to justify or conceal mirror life–enabling work:

- **Establish agreements not to pursue mirror life:** Formal or informal commitments among major scientific powers can signal shared restraint and clarify that mirror life development is outside the bounds of responsible research.
- **Expand monitoring:** Enhanced scientific monitoring can increase the likelihood of detecting early-stage mirror life–enabling research, shaping incentives by raising the perceived risk of discovery.
- **Develop verification methods:** Investment in analytical tools and inspection mechanisms can offer pathways for reducing uncertainty about ambiguous research activities. Because these methods require technical innovations that may require long lead times, we include their development in LOE 2.

- **Establish a global scientific oversight committee:** An international body, modeled on WHO or Inter-Academy mechanisms, can assess risks and provide transparent scientific guidance that reduces the potential for misinterpretation and builds collective confidence in restraint.

Collaborative implementation could include reciprocal monitoring commitments among scientifically advanced states; joint scientific reviews clarifying which research should be considered off-limits; shared development of verification benchmarks; and coordinated proposals in WHO, UN, or Organisation for Economic Co-operation and Development forums. Early progress on these levers would help create a common understanding of what constitutes concerning activity, making restraint more predictable and reducing the risk of misinterpretation.

The second set of LOE 2 levers reduces incentives and constrains technical capability:

- **Increase domestic economic controls:** Raising the economic and structural costs of pursuing mirror life in the United States is a flexible, early-stage deterrence lever that can reshape incentives without requiring criminalization or formal international agreements.
- **Restrict access to enabling materials, technologies, and knowledge:** Such actions can slow development and reinforce norms that mirror life is outside the bounds of responsible science.
- **Criminalize and prosecute mirror life development:** Domestic legislation can punish the creation of mirror organisms or research that is deemed too enabling of mirror life creation. International coordination might lead to harmonized criminal prohibitions among key states.

These levers can be implemented incrementally, often through existing domestic structures, such as patent offices, regulatory bodies, export-control agencies, and scientific publishers. Implementation with scientific superpowers could involve harmonized precursor lists, shared criteria for high-risk research, parallel restrictions on and criminalization of research to prevent regulatory arbitrage, and coordinated approaches to IP nonrecognition and publication standards.

Actors Affected

LOE 2 shapes both incentives and capabilities across the following broad set of actors:

- **Scientifically advanced nation-states:** These are most affected because their diplomatic and regulatory institutions become central for verification, agreements, and oversight.
- **Commercial entities:** These are influenced by rising direct costs, liability exposure, reduced access to materials, and fewer opportunities to capture economic benefit.
- **Academic institutions:** Such institutions are increasingly constrained by ethics review and compliance enforcement, in addition to funding conditionality.
- **Isolated or sanctioned states:** These states are somewhat directly constrained by surveillance, export controls, and diplomatic isolation. However, they are indirectly greatly constrained by scientifically advanced nation-states' collective refusal to develop the science and engineering that would enable mirror life.
- **Extremist groups and lone actors:** These are least affected directly. Their capabilities are directly limited by material controls, monitoring, and restricted institutional access. However, they are indirectly greatly constrained by scientifically advanced nation-states' collective refusal to develop the science and engineering that would enable mirror life.

Risks and Potential Pitfalls

LOE 2 levers carry several potential risks:

- **Domestic political and regulatory resistance:** Efforts to restrict scientific funding, regulate precursor materials, or impose new liability or compliance requirements might face opposition, especially if businesses perceive competitiveness or innovation risks.
- **Limited international reassurance:** Even strong U.S. domestic restraint might not fully dispel adversary suspicions that the United States maintains covert programs. However, not taking domestic action would likely worsen these misperceptions when the United States asks other nation-states to restrain themselves.
- **Geopolitical spillover:** Bilateral or multilateral agreements could be stalled or derailed by unrelated political tensions, economic disputes, or security competition. If any party suspects that other parties are seeking to preserve benefits or use mirror life governance as leverage, consensus may collapse.

LOE 2 might require reassessment if any of the following events occur:

- International monitoring is perceived as intrusive or biased or is perceived as cover for mirror life development, thus undermining trust.
- International agreements fail or trigger hedging and competitive behavior.
- Scientific discoveries reveal unexpectedly positive uses for mirror biology, or risks appear substantially lower than previously thought.
- Cost-based deterrents or export controls push activity underground or erode trust.

Under these conditions, the strategy might need to shift toward repairing legitimacy, narrowing or sequencing agreements, or re-emphasizing transparency to prevent escalation or unintended competitive dynamics.

The Role of Coercive Measures

Consistent with Principle 4, we do not include disruption or punishment as strategic levers in this report. Although such measures might appear to offer tools for deterring or halting mirror-life development, they are unlikely to be effective in practice and more likely to undermine the very cooperation on which prevention depends.

Disruptive or punitive actions—such as sanctions, cyber operations, or other coercive measures—would be difficult to apply legitimately or proportionately in the absence of a universally recognized governance framework. The perception of coercion could fracture emerging consensus, reinforce competitive narratives, and accelerate (rather than prevent) mirror life development.

Moreover, the technical and political challenges of attribution would make disruption and punishment unreliable deterrents. Mirror life development could occur in small, distributed settings, making it difficult to identify responsible parties with confidence. Premature or mistaken use of coercive tools could delegitimize broader prevention efforts and weaken the credibility of any future enforcement regime.

Disruption and punishment might become appropriate after a global consensus on mirror life risks has been achieved and legitimate multilateral mechanisms for enforcement exist, especially if science progresses to the point that restraint among scientific powers is no longer sufficient to block those who would create mirror life with the intent of doing harm. If that point is reached, coercive measures should be developed, authorized, and enforced within the same cooperative framework that made restraint possible in the first place. They should be narrowly scoped, transparently governed, and applied collectively rather than unilaterally.

At this stage, however, the sole strategic priority is to preserve the world order's capacity for collective action. Actions that rely on coercion or exclusion are out of scope.

The Role of Countermeasures and Resilience

Consistent with Principle 1, we do not include countermeasures as a strategic goal because they appear unlikely to be effective in protecting human populations, plants, animals, or the environment. Instead, these levers may have perverse consequences on efforts to halt the development of mirror life. The pursuit of countermeasures could be seen as condoning or enabling mirror life development. Even exploratory efforts might be misinterpreted as evidence of an active program. Worse, activities framed as countermeasures might be indistinguishable from actual mirror life development programs, creating suspicion and inadvertently triggering competitive dynamics.

However, exploratory countermeasure and resilience research activities might be appropriate after consensus is achieved on risks among scientifically advanced nation-states or in limited other circumstances. If mirror life countermeasure development is contemplated, it should

- avoid advancing mirror life creation, either directly or indirectly
- be confined to nonmaterial research (tabletop exercises, modeling studies, or theoretical assessments), and there should be no physical materials or systems being developed, tested, or stockpiled
- be conducted transparently, potentially by international organizations, and there should be full international visibility
- be subject to verification and shared oversight, ideally through a multilateral regime
- offer any discoveries or benefits to all parties globally, including those not participating in countermeasure development efforts.

Importantly, if mirror life were ever created or released, the calculus would change. At that point, prohibitions against research would need exceptions to allow scientific, public health, law enforcement, and national security communities to detect, collect, study, and possibly reproduce mirror organisms to respond. Yet even then, countermeasure development would face a grim race: between creating defenses and the spread of mirror organisms through the biosphere. If initial mirror organisms proved fragile and harmless, developing countermeasures that unintentionally accelerated progress toward more-robust forms would be a grave mistake.

Conclusion

Preventing the creation of mirror life is a global challenge. It requires a strategy of early action and sustained coordination even in the face of scientific uncertainty and geopolitical complexity. The LOEs outlined in this chapter are mutually reinforcing components of a broader prevention-first approach and are summarized in Table 6.1.

The logic is straightforward. If mirror life poses even a small chance of catastrophic harm, then acting before technical feasibility emerges is essential. If mirror life anywhere is a threat to life everywhere, then all actors must be deterred. However, the silver lining is that the path to universal restraint runs through the restraint of the capable. If collaboration among major scientific powers is prioritized, prevention becomes possible.

Successful implementation will require navigating real limits: incomplete scientific knowledge, contested geopolitical relationships, uneven regulatory environments, and actors who may not share the same values or incentives. The United States cannot, on its own, control whether mirror life is ever pursued, but it can play a central and collaborative role in shaping the conditions that make restraint the widely accepted and most rational choice. The strategy presented in this report is not a prediction of what will occur but rather an attempt to outline a feasible path that reduces the likelihood that mirror life is ever created. If pursued with clarity and humility, this path might help ensure that mirror life remains a scientific possibility that the world collectively chooses not to realize.

Abbreviations

AI	artificial intelligence
BWC	Biological Weapons Convention
CFC	chlorofluorocarbon
DNA	deoxyribonucleic acid
ENMOD	Environmental Modification Convention
GDP	gross domestic product
IP	intellectual property
ISIS	Islamic State of Iraq and Syria
LOE	line of effort
MBDF	Mirror Biology Dialogues Fund
NATO	North Atlantic Treaty Organization
NSF	National Science Foundation
PPE	personal protective equipment
PSI	Proliferation Security Initiative
R&D	research and development
RNA	ribonucleic acid
UN	United Nations
UNESCO	United Nations Educational, Scientific and Cultural Organization
WHO	World Health Organization
WMD	weapon of mass destruction

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